

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck



Deck and window side

GENERAL NOTES

- All dimensions are to framing unless noted. Rough-sawn boards are full dimension; verify actual sizes delivered by the mill and update the stock table before cutting.
- All lumber within 6" of grade, and all skids, rims, joists, posts and bottom plates: pressure treated. Skids UC4B (ground contact).
- Fasteners in contact with treated lumber: hot-dip galvanized (ASTM A153) or stainless. Siding: 8d ring-shank HDG, battens: 10d ring-shank HDG nailed through the gap only, so boards can move.
- Board-and-batten boards are nailed to horizontal blocking at 24" max. and to plates. Leave a 1/2" gap between boards for shrinkage of air-dried lumber.
- North Carolina: privies fall under the state on-site wastewater rules administered by the county Environmental Health office. Confirm the permit, the setbacks (commonly 100' from a private well, plus streams and property lines) and the seasonal high water table before digging.
- NC Piedmont red clay drains slowly, so the pit works largely as a storage vault. Add a scoop of sawdust, wood ash or dry leaf litter after each use, and keep roof and surface water away from the pit.
- Subterranean termites are active throughout the Piedmont. Keep untreated wood at least 6" above grade (the siding line is set by `building.siding.min_clearance_to_grade`), use UC4B for skids and posts, and inspect the skids every year.
- Wooded site: keep out from under dead or leaning trees, cut roots cleanly when augering the pit, and leave the pad crowned so leaf-litter runoff drains away.
- The pipe is set plumb and level-topped, backfilled with compacted native soil in 6" lifts. The pit has an open bottom; do not bed it in concrete.
- Screening structural checks are on A-602; the cut list and purchase list are on A-601. This is not a stamped engineered design.
- Relocation: turn off the branch breaker at the pedestal, open the disconnect, unhook the feeder at the entry box, remove the back panel and flange, then pull the sled off the pit from the front skid tow holes. See sheets A-502 and E-101.
- Electrical: one 120 V, 20 A GFCI branch circuit from the house panel, installed to the NEC as adopted in NC and inspected by Rockingham County. See E-101.

SHEET INDEX

SHEET	TITLE
G-001	Cover, sheet index & general notes
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A-102	Roof plan & roof framing plan
S-101	Foundation & floor framing plan
A-201	Exterior elevations: left & front
A-202	Exterior elevations: right & back
A-301	Section A: longitudinal through pit
A-302	Sections B & C: transverse
S-201	Left (low) side wall framing
S-202	Right (high) side wall framing
S-203	End wall framing elevations
A-501	Details: pit & flange, skid, rafter
A-502	Installation & relocation sequence
A-503	Details: removable panel, door, bench
E-101	Electrical plan
A-601	Cut list & material schedules
A-602	Design checks, parameters & render setup
R-101	Renderings

DESIGN BASIS

ITEM	VALUE
Ground snow pg	15 psf
Roof snow pf	12.0 psf
Roof dead	6 psf
Floor / deck live	40 / 40 psf
Bench occupant	300 lb point
Est. empty weight	2,340 lb
Tow force on dirt	1,170-1,638 lb
Usable pit volume	55 gal
Est. pit service life	12.7 yr

DESIGN CHECKS

33 pass, 0 fail, 8 info (A-602)

HOW TO REVISE

Edit the JSON (dimensions, stock, materials, lighting, cameras) and run `python -m privy build`. Sheets, cut list, checks and renderings regenerate from it.

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

SITE	Wooded lot, Rockingham County, NC (Piedmont)
OWNER	—
DESIGN	—

KEY DIMENSIONS

Footprint (framing)	5'-0" x 7'-0"
Deck	5'-0" x 6'-0"
Skids	14'-0" long
Roof	3:12 single slope, metal
Pit	30" ID HDPE, 3'-0" deep
Height to roof	10'-5 9/16"

PARAMETRIC SOURCE

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generated 2026-09-28

REV	DATE	DESCRIPTION
A	2026-09-28	Issued for review

SHEET TITLE

COVER, SHEET INDEX & GENERAL NOTES

SCALE: AS NOTED

DATE: 2026-09-28

SHEET 1 OF 18

G-001

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

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SHEET TITLE

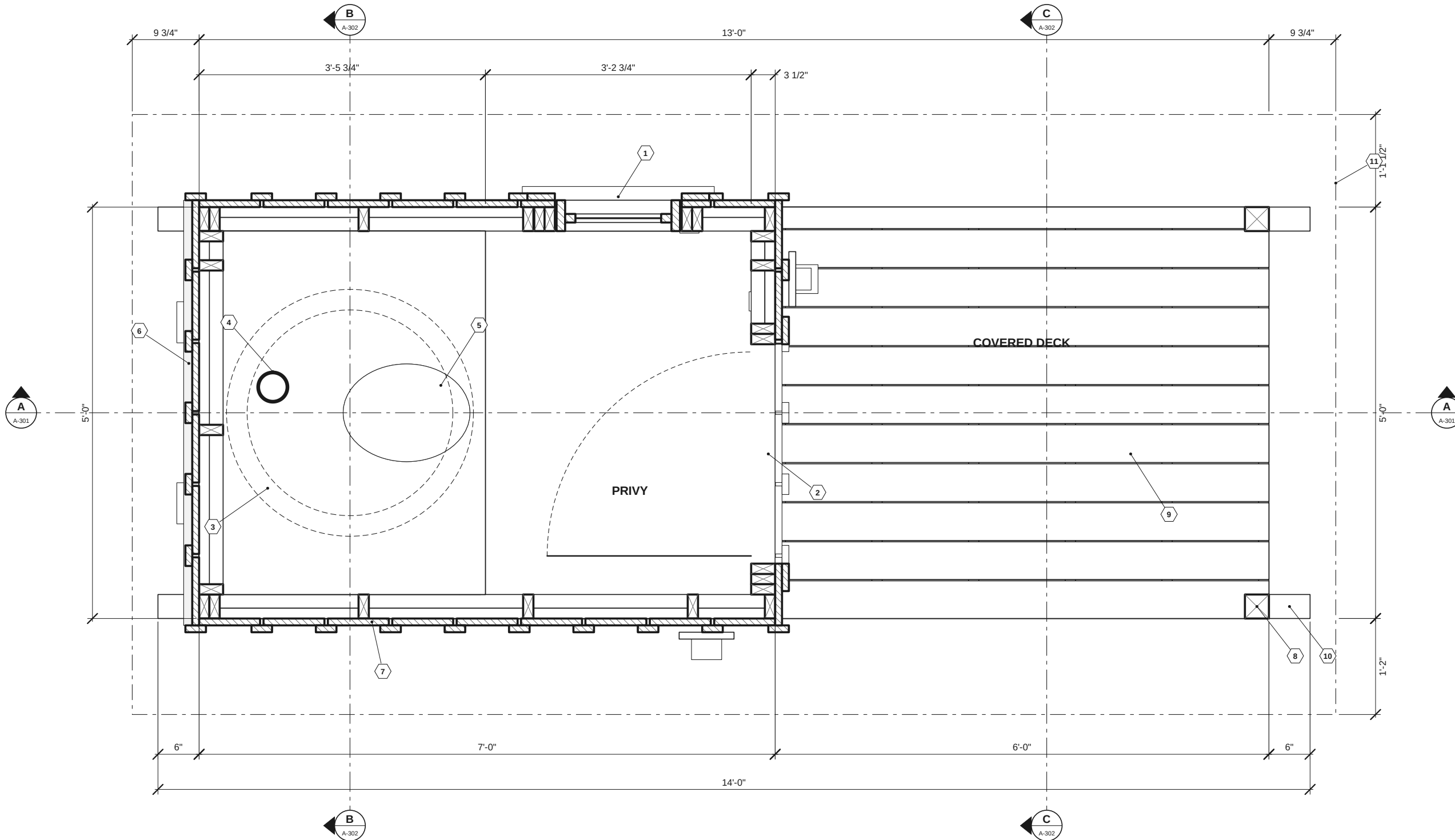
FLOOR PLAN

SCALE: AS NOTED

DATE: 2026-09-28

SHEET 2 OF 18

A-101



1 FLOOR PLAN

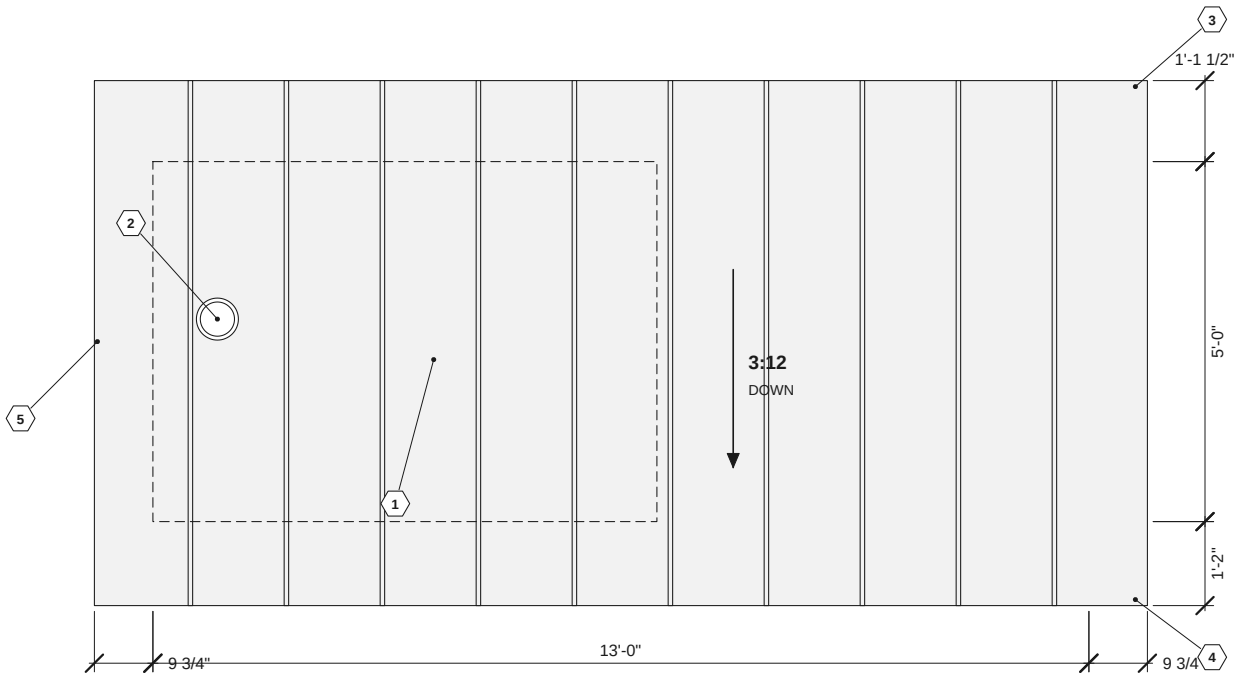
SCALE: 3/4" = 1'-0"

KEYNOTES

- | | |
|--|--|
| 1 W1 window: 18" x 27" double-hung, sill at 4'-8" above floor (see schedule) | 6 Removable back panel below bench (see A-202, A-501) |
| 2 D1: Fiberglass prehung exterior door, 2'-6" x 6'-8", in-swing, smooth 2-panel, 4-9/16" jamb. Keyed lever + deadbolt, factory hinges, adjustable sill, weatherstrip; exterior casing rough-sawn 1x4 in-swing; the leaf folds against the side wall. | 7 2x4 studs @ 24" o.c.; 1x10 rough-sawn boards + 1x3 battens |
| 3 30" HDPE pit below the bench (hidden) | 8 4x4 PT post on skid; knee brace to beam above |
| 4 4" PVC vent from flange up through roof | 9 5/4x6 PT decking, 3/16" gaps, notched at posts |
| 5 Toilet seat on 3/4" plywood bench, top at 1'-5" above floor | 10 4x6 PT skid, chamfered ends with tow hole |
| | 11 Line of roof overhang above |

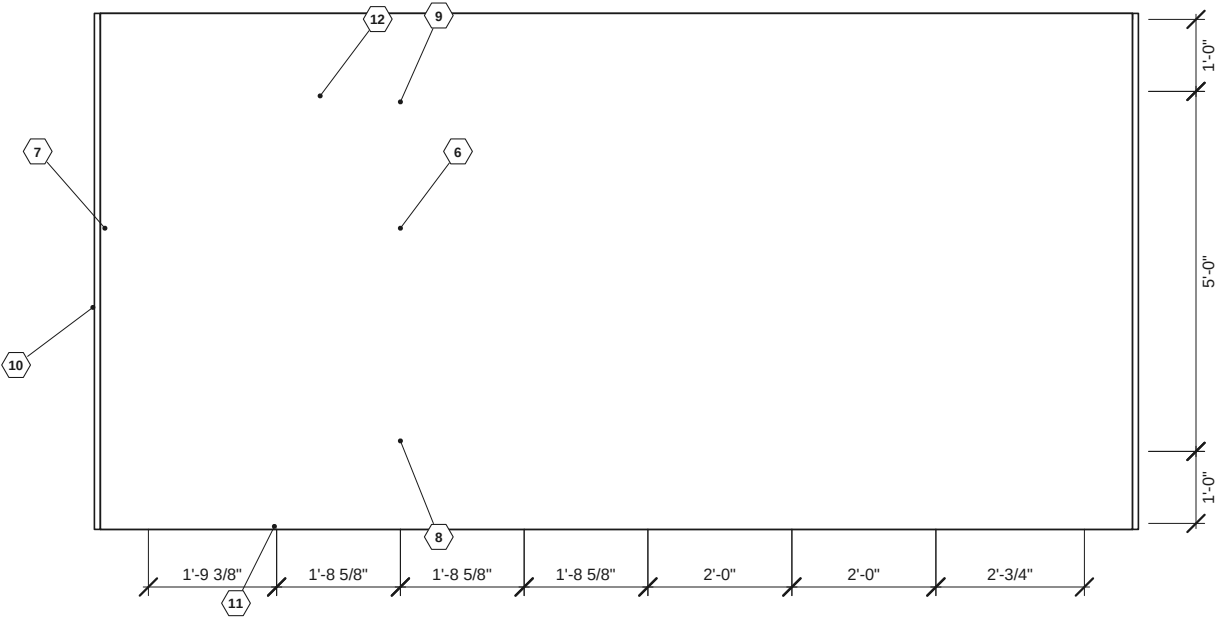
DOOR & WINDOW SCHEDULE

TAG	TYPE	R.O.	REMARKS
D1	Door	32" x 82 1/2"	Fiberglass prehung exterior door, 2'-6" x 6'-8", in-swing, smooth 2-panel, 4-9/16" jamb. Keyed lever + deadbolt, factory hinges, adjustable sill, weatherstrip; exterior casing rough-sawn 1x4; swing in, hinge left (from outside)
W1	Window	18 1/2" x 27 1/2"	Small single-glazed double-hung shed window, wood or vinyl, 18" x 27" unit; sill +4'-8" AFF



1 ROOF PLAN

SCALE: 3/8" = 1'-0"



2 ROOF FRAMING PLAN

SCALE: 3/8" = 1'-0"

ROOF NOTES

- Single-slope roof drains to the low (left) side, away from the door and the pit.
- Snow: pg = 15 psf, pf = 12.0 psf (ASCE 7, unheated, Risk Cat. I).
- Every rafter: hurricane tie to beam. The whole roof is part of the towed assembly.
- Keep 10' from dead limbs; a metal roof tolerates small branch and cone falls.

KEYNOTES

- 24 ga Galvalume snap-lock standing seam, 16" panels, 1" seams, over synthetic underlayment; 5/8" CDX plywood roof deck on the rafters
- 4" vent through EPDM roof boot, screened cap 18" above roof
- High edge: outside foam closure + peak (shed) flashing
- Low eave: sheets overhang fascia 1 1/2"; inside foam closure
- Rake: sheets overhang 1x6 barge board 1"; optional rake trim
- 2x6 rafters @ 24" o.c. max, birdsmouth on both beams, plumb-cut tails (see A-501)
- 2x6 fly rafter at each rake, hung from the sheathing and fascia
- 4x6 beam, low side, continuous from back wall to deck post
- 4x6 beam, high side, continuous from back wall to deck post
- 1x6 rough-sawn barge board
- 1x6 rough-sawn fascia on rafter tails
- 2x6 frieze blocks between rafters over building walls

SKID-MOUNTED PIT PRIVY

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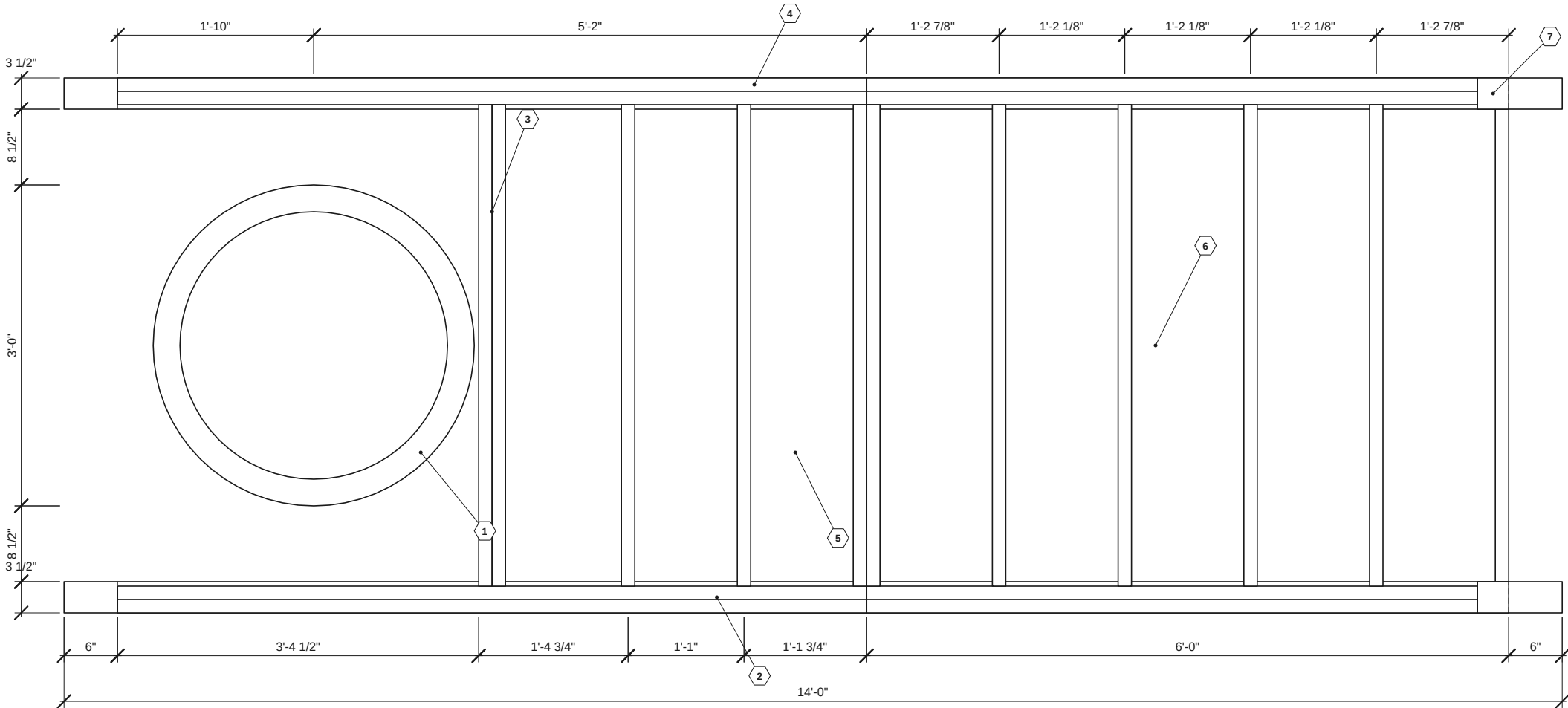
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SHEET TITLE

ROOF PLAN & ROOF FRAMING PLAN

SCALE: AS NOTED
DATE: 2026-09-28
SHEET 3 OF 18

A-102



1 FOUNDATION & FLOOR FRAMING PLAN
SCALE: 3/4" = 1'-0"

KEYNOTES

- 1** 30" dual-wall HDPE pipe (36" OD) set before the privy arrives; passes between the skids with 8 1/2" clear each side
- 2** 4x6 PT UC4B skid on edge, full length of privy + deck, chamfered both ends, tow hole each end (A-501)
- 3** (2) 2x6 PT bench-front joist; no floor framing behind it (pipe path)
- 4** (2) 2x6 PT rim joists on the skid, screwed down with 1/4" x 6" structural screws @ 16" o.c.
- 5** 2x6 PT floor joists @ 16" max, joist hangers; 3/4" PT plywood floor
- 6** 2x4 PT deck joists @ 16" o.c. between (2) 2x4 PT rims, hangers
- 7** 4x4 PT post on skid, galv. post base, (2) 1/2" bolts to rim

FOUNDATION NOTES

- No foundation: the privy and deck ride on two continuous skids (a sled) on a level dirt pad.
- Grade the skid paths flat and compact them; shim with PT blocks or flat stones, never with the pipe.
- Red clay (NC Piedmont) holds water: crown the pad slightly so it drains away from the pit and skids.
- Tow only from both skid ends at once (bridle), with the back panel installed and lagged.

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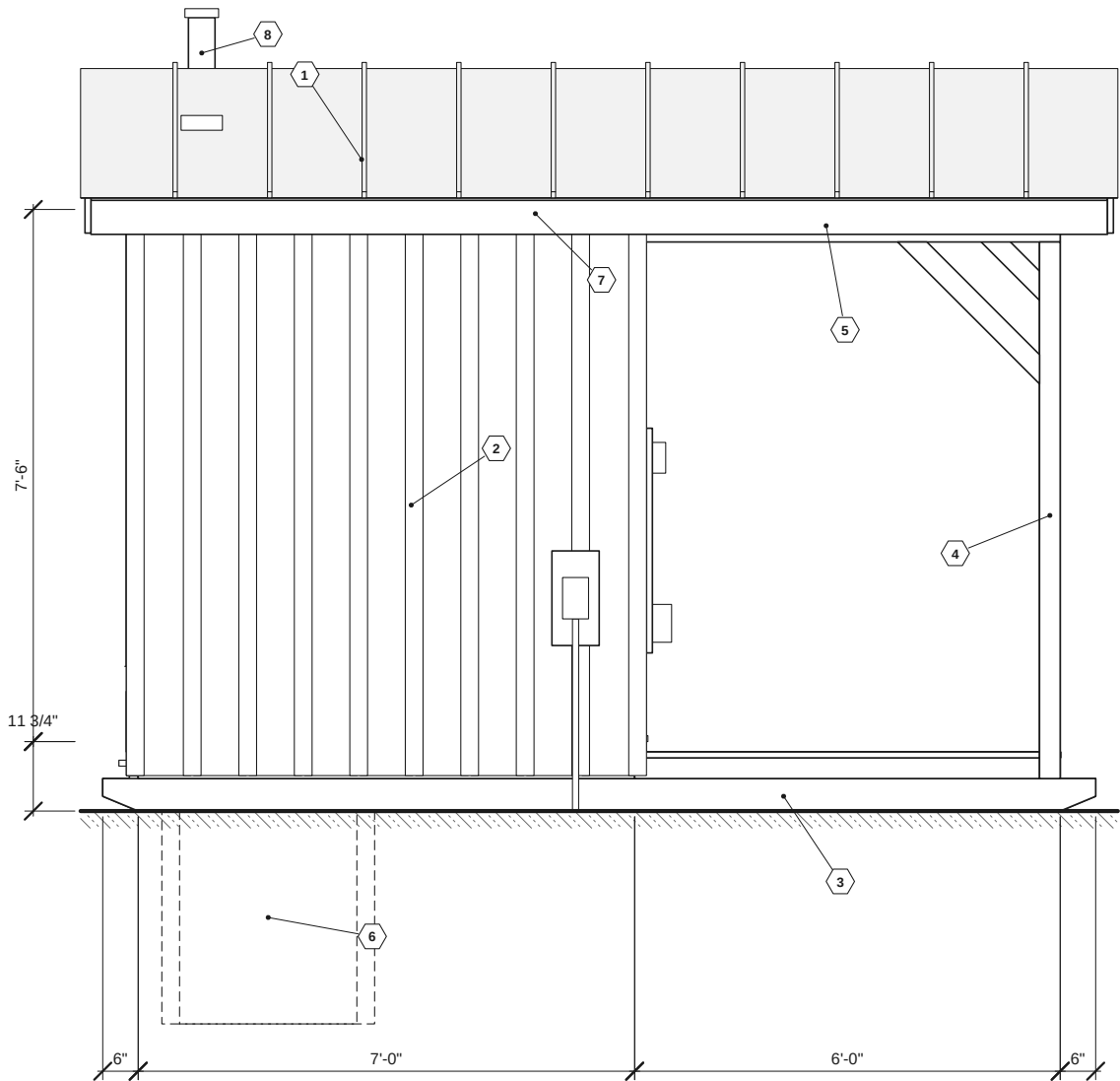
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FOUNDATION & FLOOR FRAMING PLAN

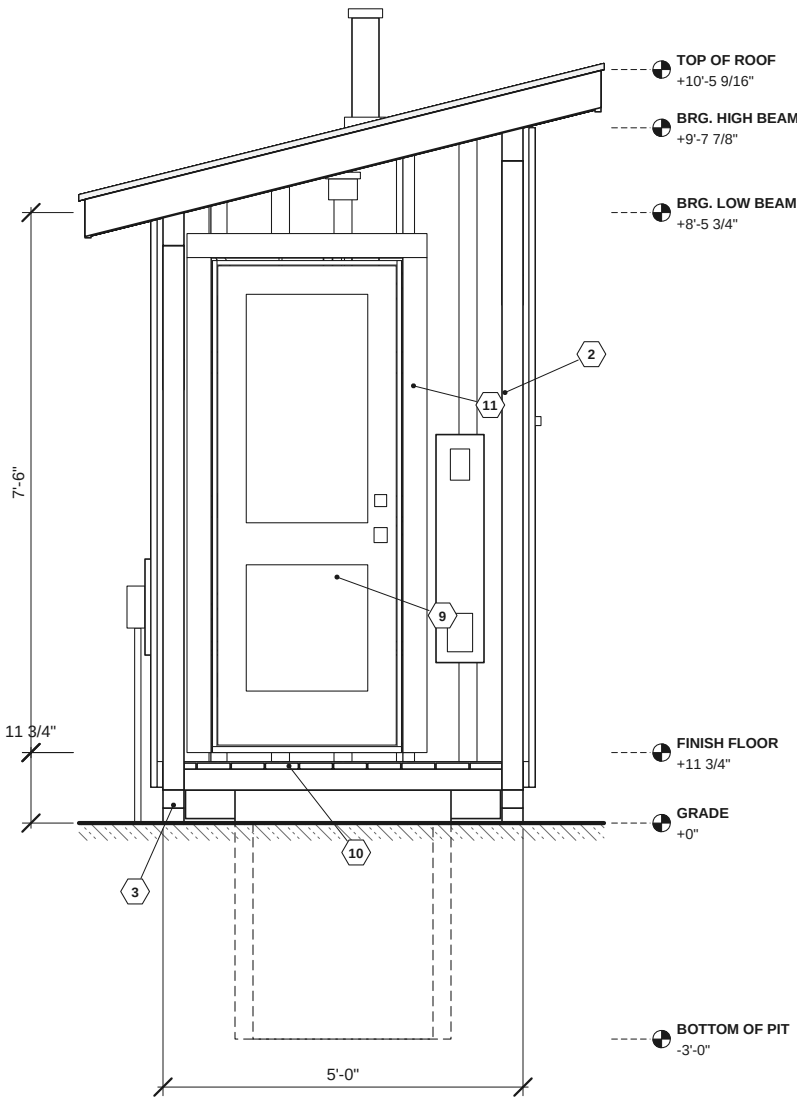
SCALE: AS NOTED
DATE: 2026-09-28
SHEET 4 OF 18

S-101



1 LEFT (LOW / EAVE SIDE) ELEVATION

SCALE: 3/8" = 1'-0"



2 FRONT (DOOR / DECK) ELEVATION

SCALE: 3/8" = 1'-0"

KEYNOTES

- 24 ga Galvalume snap-lock standing seam, 16" panels, 1" seams, over synthetic underlayment; 5/8" CDX plywood roof deck on the rafters; 3:12
- 1x10 rough-sawn boards, 1/2" gaps, 1x3 battens (board-and-batten)
- 4x6 PT skid on edge, chamfered ends
- 4x4 PT post with 4x4 knee brace to beam
- 4x6 beam, continuous over deck

- 30" HDPE pit, 3'-0" below grade (hidden)
- 1x6 rough-sawn fascia on plumb-cut rafter tails
- 4" PVC vent, screened cap
- D1: Fiberglass prehung exterior door, 2'-6" x 6'-8", in-swing, smooth 2-panel, 4-9/16" jamb. Keyed lever + deadbolt; factory hinges, adjustable sill, weatherstrip; exterior casing rough-sawn 1x4
- 5/4x6 PT deck on 2x4 PT joists

- 1x4 rough-sawn door casing

SKID-MOUNTED PIT PRIVY

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DESIGN	—

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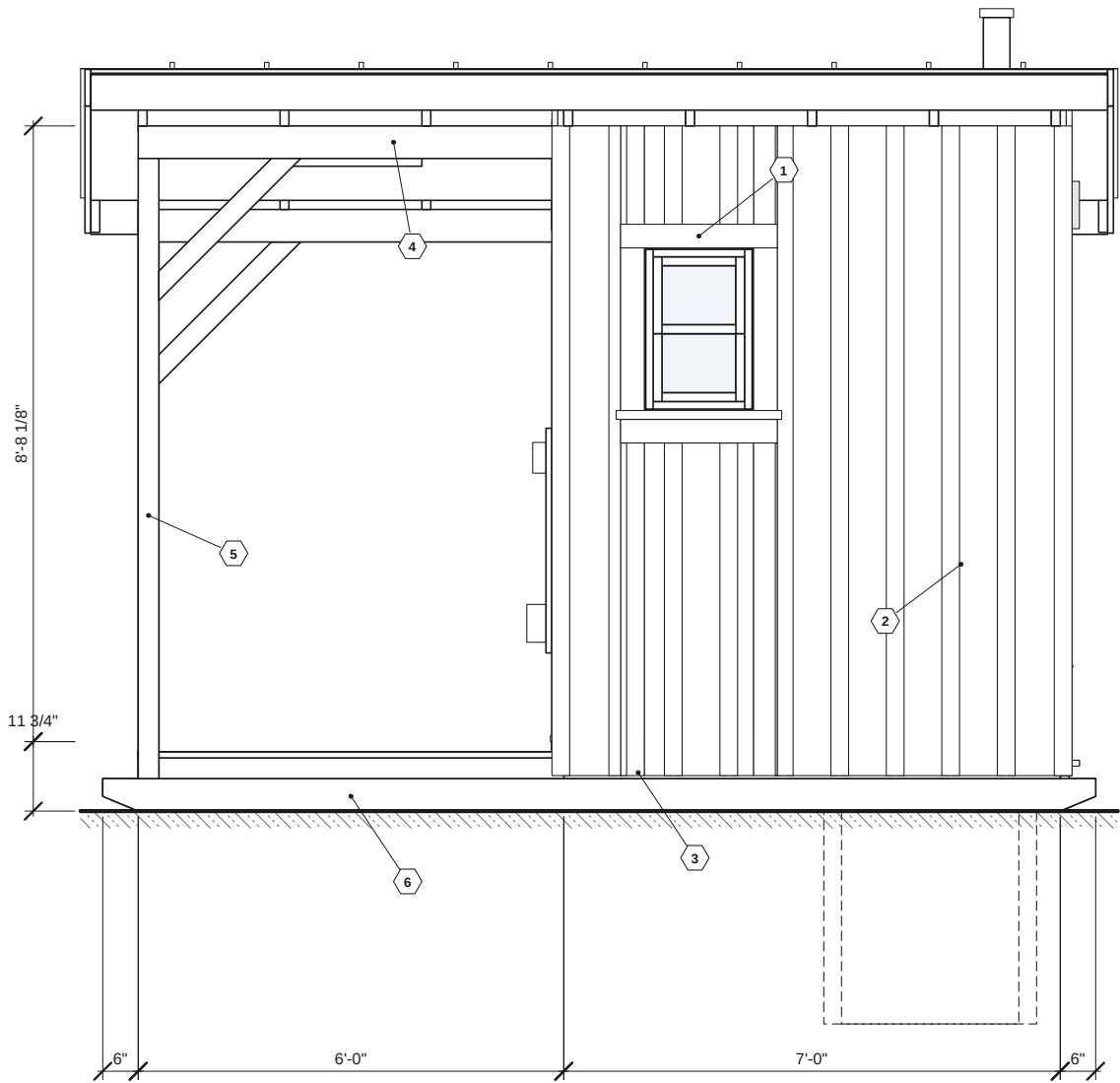
EXTERIOR ELEVATIONS: LEFT & FRONT

SCALE: AS NOTED

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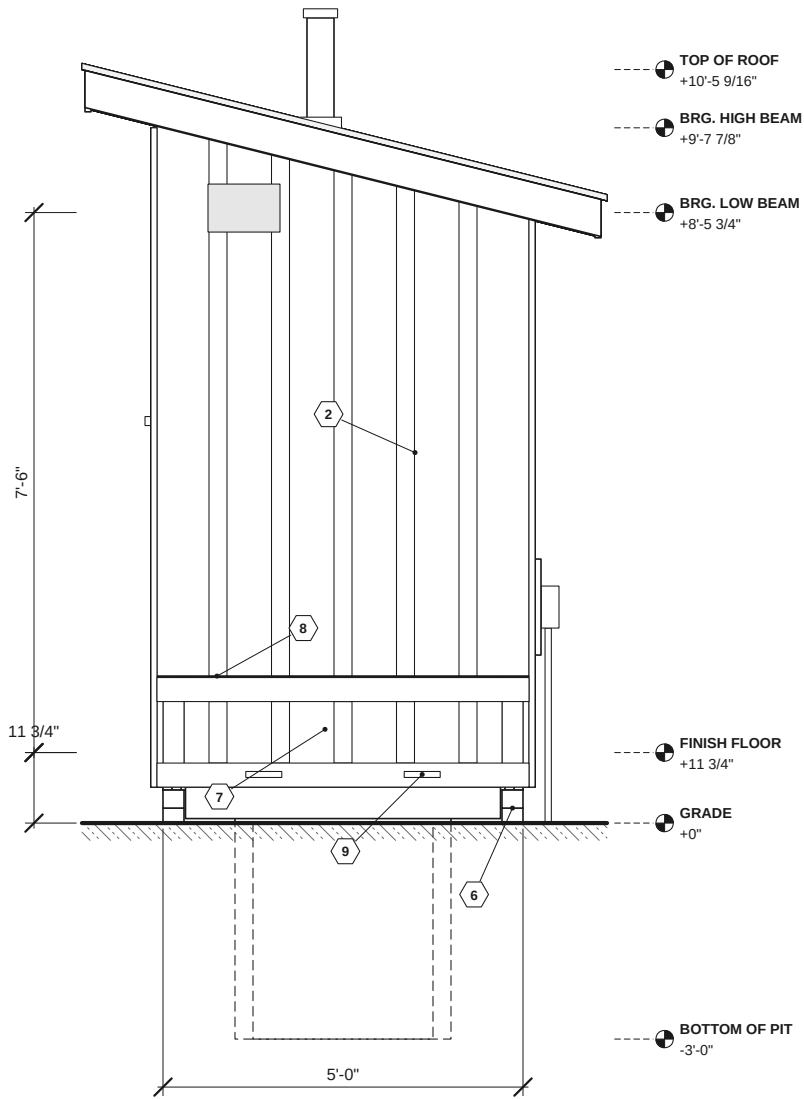
SHEET 5 OF 18

A-201



1 RIGHT (HIGH / WINDOW SIDE) ELEVATION

SCALE: 3/8" = 1'-0"



2 BACK (REMOVABLE PANEL) ELEVATION

SCALE: 3/8" = 1'-0"

KEYNOTES

- 1 W1 double-hung window, 1x4 rough-sawn casing, sill and apron
- 2 1x10 rough-sawn boards, 1/2" gaps, 1x3 battens (board-and-batten)
- 3 Siding stops 6" above grade (IRC R317.1 decay / termite clearance); PT rim exposed below
- 4 4x6 beam, continuous over deck
- 5 4x4 PT post with 4x4 knee brace to beam

- 6 4x6 PT skid on edge, chamfered ends
- 7 Removable panel: boards on 1x4 rails, (10) 1/4" lags; remove to slide the privy over the pit, then re-install
- 8 Galvanized Z-flashing at panel head
- 9 Pull handles

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

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DESIGN	—

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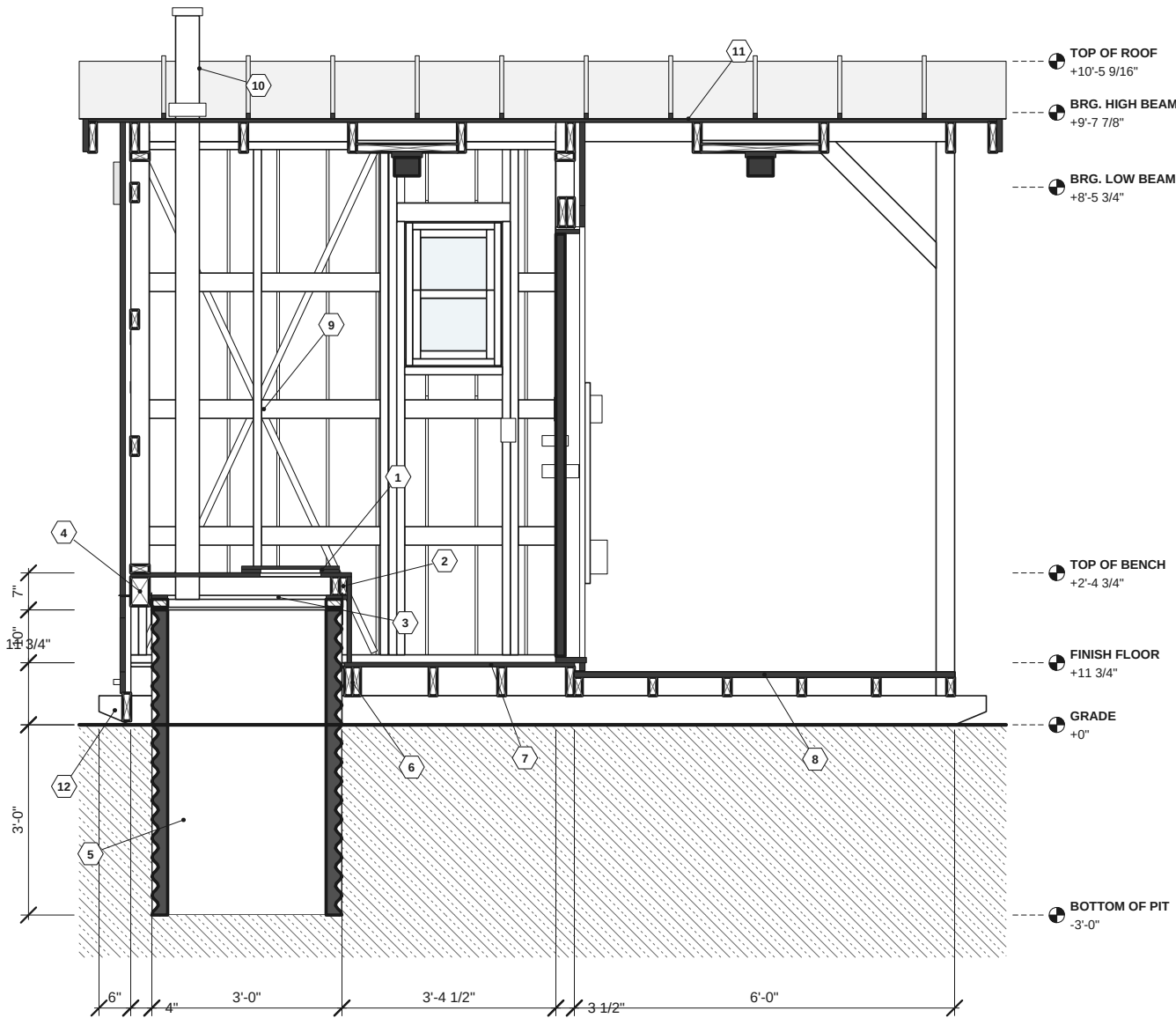
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SHEET TITLE

EXTERIOR ELEVATIONS:
RIGHT & BACK

SCALE: AS NOTED
DATE: 2026-09-28
SHEET 6 OF 18

A-202



KEYNOTES

- 1 Toilet seat with lid on 3/4" BC plywood bench top, sealed
- 2 2x4 bench framing: front rail, joists, side ledgers; 3/4" plywood front panel
- 3 Flange: 3/4" PT plywood plate with 30" opening + 2-ply collar; EPDM skirt with SS band clamps (A-501)
- 4 4x6 PT bench header over the panel opening
- 5 30" dual-wall HDPE pipe, open bottom, compacted native backfill
- 6 (2) 2x6 PT bench-front joist; open behind for the pipe
- 7 2x6 PT joists @ 16", 3/4" PT plywood floor
- 8 5/4x6 PT decking on 2x4 PT joists @ 16" o.c.
- 9 2x4 studs @ 24" o.c., flat blocking @ 24", let-in strap bracing
- 10 4" PVC vent, screened cap
- 11 2x6 rafters @ 24" max; 24 ga Galvalume snap-lock standing seam, 16" panels, 1" seams, over synthetic underlayment; 5/8" CDX plywood roof deck on the rafters
- 12 4x6 PT skid beyond

SKID-MOUNTED PIT
PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

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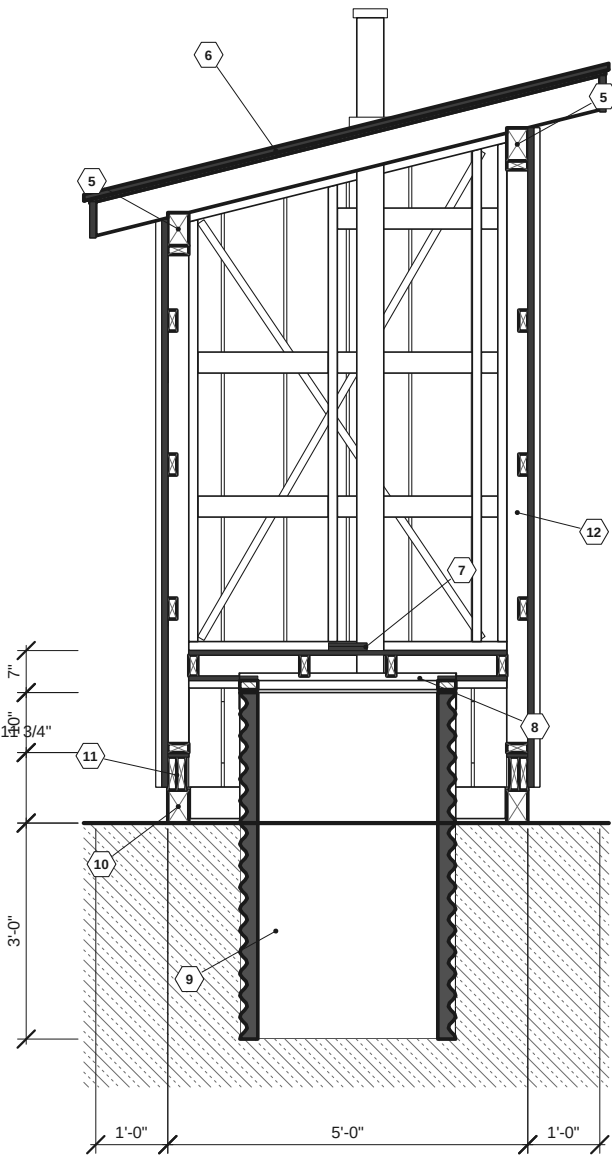
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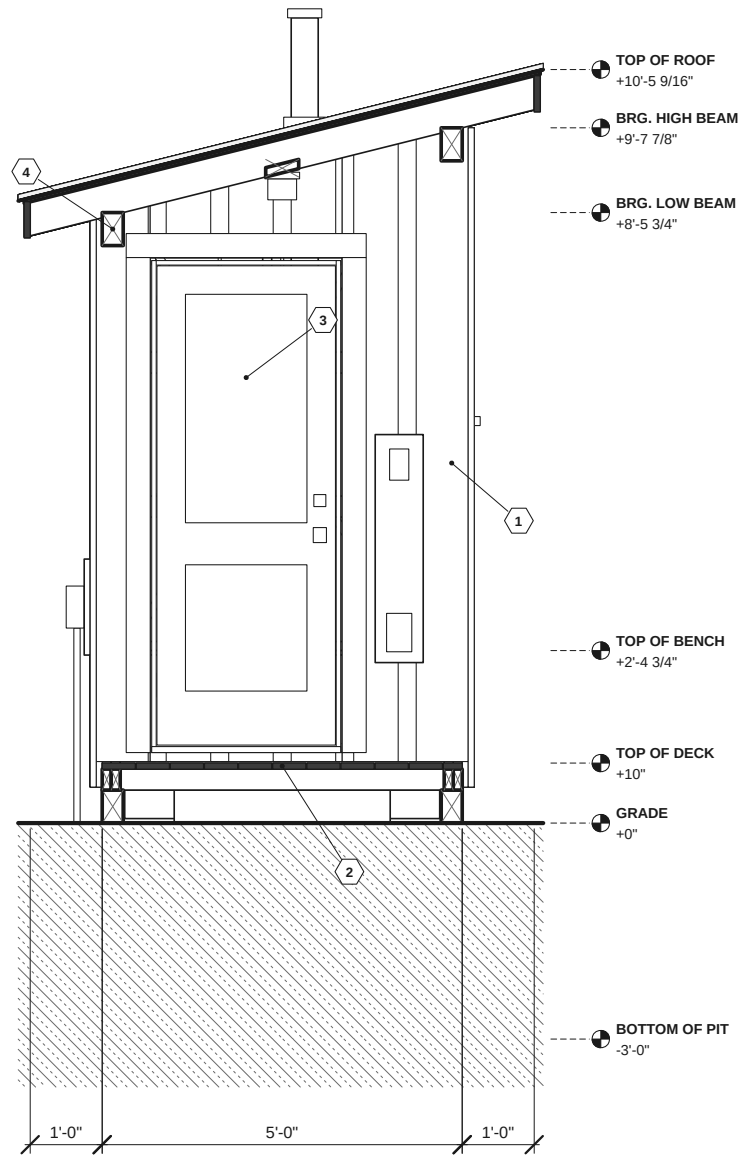
SECTION A: LONGITUDINAL
THROUGH PIT

SCALE: AS NOTED
DATE: 2026-09-28
SHEET 7 OF 18

A-301



1 SECTION B: TRANSVERSE THROUGH PIT
SCALE: 3/8" = 1'-0"



2 SECTION C: THROUGH THE DECK
SCALE: 3/8" = 1'-0"

KEYNOTES

- 4x4 PT post on skid, galv. base, (2) 1/2" bolts to deck rim
- 5/4x6 PT decking on 2x4 PT joists @ 16" between (2) 2x4 PT rims
- D1 door beyond (see A-503)
- 4x6 beam on post, 4x4 knee brace (beyond)
- 4x6 beam on 2x4 top plate, birdsmouth rafter seat
- 2x6 rafters @ 24" max; 24 ga Galvalume snap-lock standing seam, 16" panels, 1" seams, over synthetic underlayment; 5/8" CDX plywood roof deck on the rafters
- Toilet seat with lid on 3/4" BC plywood bench top, sealed
- Flange: 3/4" PT plywood plate with 30" opening + 2-ply collar; EPDM skirt with SS band clamps (A-501)
- 30" dual-wall HDPE pipe, open bottom, compacted native backfill
- 4x6 PT skid on edge
- (2) 2x6 PT rim joists
- 2x4 studs @ 24" o.c., flat blocking @ 24", let-in strap bracing

SECTION NOTES

- The pipe top (+1'-9 3/4") sits 3/4" below the bench header so the privy can slide over it; the flange closes the gap afterwards.
- The rafters bear on 4x6 beams at +8'-5 3/4" (low) and +9'-7 7/8" (high); the front and back walls are non-bearing.
- Everything above the flange plate is removable with the building; the pipe stays in the ground.

SKID-MOUNTED PIT PRIVY

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SHEET TITLE

SECTIONS B & C: TRANSVERSE

SCALE: AS NOTED
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SHEET 8 OF 18

A-302



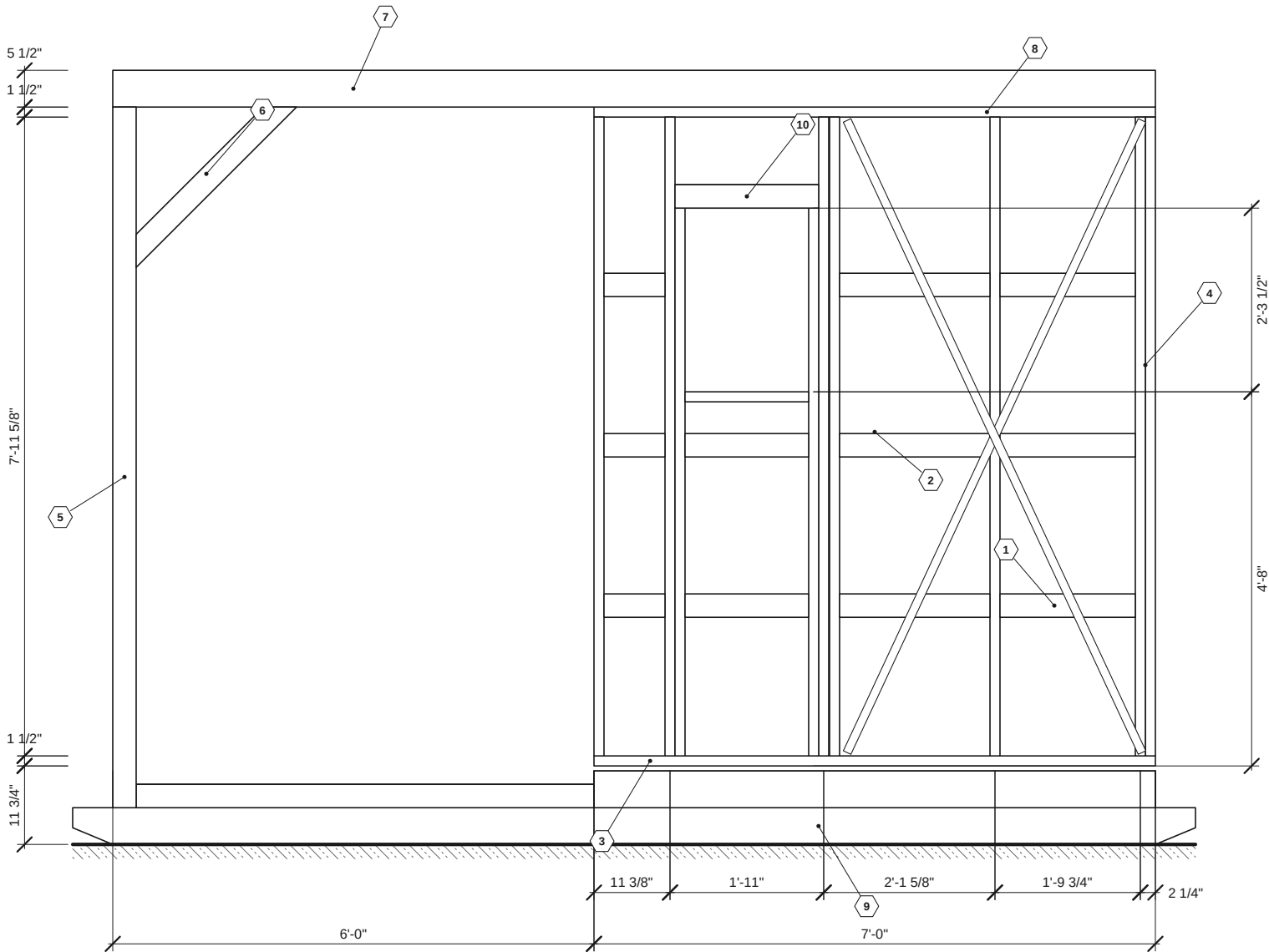
1	2x4 flat blocking @ 24" o.c., flush to outside face (siding nailers)	9	4x6 PT skid and (2) 2x6 PT rim joists
2	Let-in steel wall-bracing strap, X in the solid bay; nail at each stud and plate		
3	PT 2x4 bottom plate on 3/4" PT plywood		
4	Doubled corner studs at the back: bench-header hangers		
5	4x4 PT post on skid		
6	4x4 PT knee brace, 45 deg, 24" legs		
7	4x6 beam on top plate, continuous to the post		
8	Single 2x4 top plate (beam above acts as second plate)		

1. 2x4 SPF #2 studs @ 24" o.c.; the side walls carry the roof via one top plate plus the 4x6 beam.
2. Toe-screw and strap the beam to each corner stud pack and to the post.
3. Studs differ in length between the low and high walls; cut from 92-5/8" precuts where possible.

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SHEET 9 OF 18

S-201



SKID-MOUNTED PIT PRIVY

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KEYNOTES

- 2x4 flat blocking @ 24" o.c., flush to outside face (siding nailers)
- Let-in steel wall-bracing strap, X in the solid bay; nail at each stud and plate
- PT 2x4 bottom plate on 3/4" PT plywood
- Doubled corner studs at the back: bench-header hangers
- 4x4 PT post on skid
- 4x4 PT knee brace, 45 deg, 24" legs
- 4x6 beam on top plate, continuous to the post
- Single 2x4 top plate (beam above acts as second plate)

- 4x6 PT skid and (2) 2x6 PT rim joists
- (2) 2x4 header over W1, R.O. 18 1/2" x 27 1/2"; king + jack studs; flat 2x4 rough sill

FRAMING NOTES

- 2x4 SPF #2 studs @ 24" o.c.; the side walls carry the roof via one top plate plus the 4x6 beam.
- Toe-screw and strap the beam to each corner stud pack and to the post.
- Studs differ in length between the low and high walls; cut from 92-5/8" precuts where possible.

SHEET TITLE

RIGHT (HIGH) SIDE WALL FRAMING

SCALE: AS NOTED

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SHEET 10 OF 18

S-202

KEYNOTES

- 1

(2) 2x6 header over D1, R.O. 32" x 82 1/2"; king + jack studs
- 2

Bottom plate cut out at door after framing
- 3

Sloped 2x4 top plate under end rafter, 3:12
- 4

2x4 flat blocking @ 24" (siding nailers)
- 5

4x6 PT bench header, full interior width, face-mount hangers to doubled corner studs; removable panel opening below
- 6

2x4 PT bottom plate on bench top over header

END WALL NOTES

- 1

End walls are non-bearing: the rafters span side to side between the 4x6 beams.
- 2

The back wall has no bottom plate below the bench header. With the panel off, the bench box and the header brace the back; keep the panel lagged whenever the privy is moved.
- 3

The front wall is too narrow for strap bracing beside the door; the roof diaphragm and the floor carry racking.

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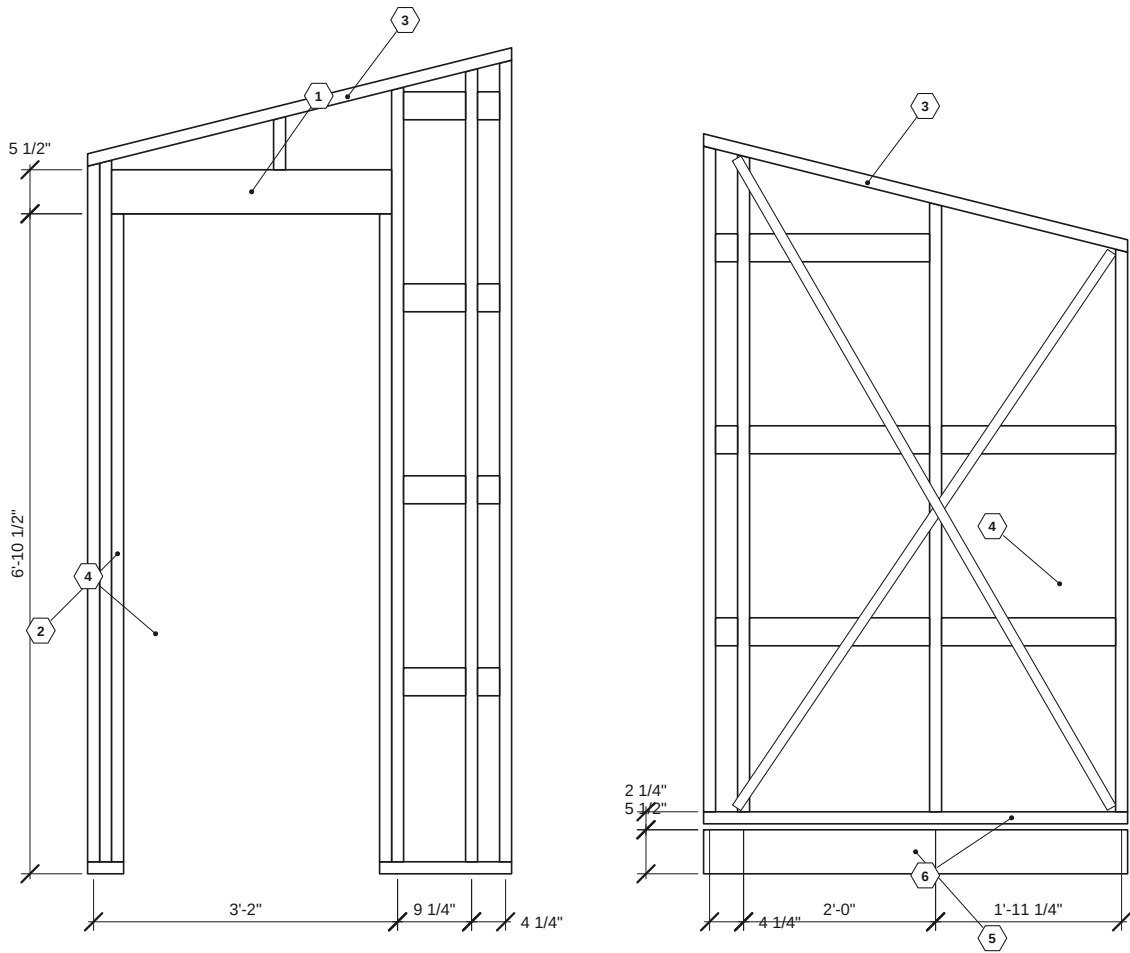
REV	DATE	DESCRIPTION
A	2026-09-28	Issued for review

SHEET TITLE

END WALL FRAMING ELEVATIONS

SCALE: AS NOTED
DATE: 2026-09-28
SHEET 11 OF 18

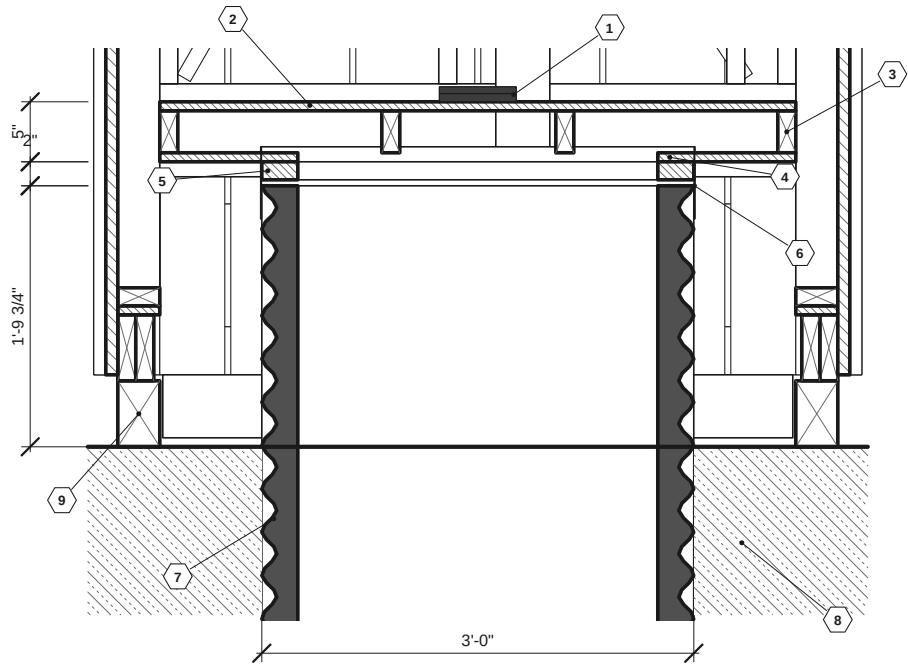
S-203



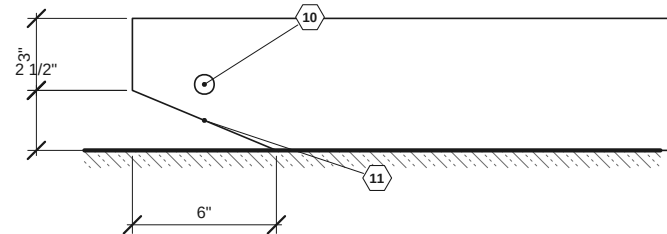
1 FRONT WALL FRAMING (FROM OUTSIDE) 2 BACK WALL FRAMING (FROM OUTSIDE)

SCALE: 1/2" = 1'-0"

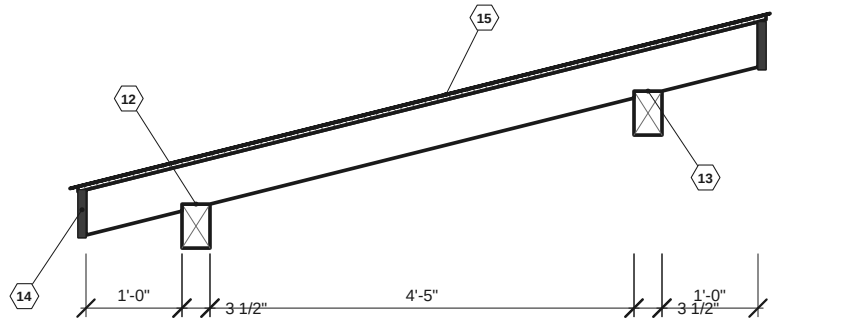
SCALE: 1/2" = 1'-0"



1 PIT & FLANGE DETAIL (SECTION B)
SCALE: 3/4" = 1'-0"



2 SKID END (BOTH ENDS)
SCALE: 1-1/2" = 1'-0"



3 RAFTER LAYOUT (TYPICAL) - 7'-2 9/16" 2X6
SCALE: 1/2" = 1'-0"

KEYNOTES

- 1 Toilet seat, stainless hinge bolts through bench top
- 2 3/4" BC plywood bench top, sealed; hole cut to the seat template
- 3 2x4 bench side ledger screwed to studs
- 4 Flange plate: 3/4" PT plywood with 30" hole; slid in from the back after the privy is placed; (12) #10 x 3" SS screws up into bench framing
- 5 Collar: 2 plies 3/4" PT plywood rings (half rings glued & screwed), OD = pipe OD
- 6 EPDM skirt over the gap, (2) stainless band clamps; foam gasket under collar
- 7 30" dual-wall HDPE pipe, cut square, plumb
- 8 Backfill: native soil in 6" lifts, hand tamped; no concrete
- 9 4x6 PT skid, clear of pipe
- 10 13/16" tow hole 3" from end, for a 3/4" shackle
- 11 Chamfer 6" x 2 1/2" on the bottom edge, both ends, so the sled rides up over roots and ruts
- 12 Birdsmouth: 3 1/2" seat, 7/8" heel; hurricane tie each rafter
- 13 Birdsmouth at high beam, heel cut on the inside face
- 14 Plumb-cut tail, 1x6 rough-sawn fascia
- 15 24 ga Galvalume snap-lock standing seam, 16" panels, 1" seams, over synthetic underlayment; 5/8" CDX plywood roof deck on the rafters; 3:12

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

SITE	Wooded lot, Rockingham County, NC (Piedmont)
OWNER	—
DESIGN	—

KEY DIMENSIONS

Footprint (framing)	5'-0" x 7'-0"
Deck	5'-0" x 6'-0"
Skids	14'-0" long
Roof	3:12 single slope, metal
Pit	30" ID HDPE, 3'-0" deep
Height to roof	10'-5 9/16"

PARAMETRIC SOURCE

design/privy.json
model hash d81dcef27b
generated 2026-09-28

REV	DATE	DESCRIPTION
A	2026-09-28	Issued for review

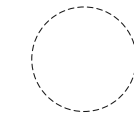
SHEET TITLE

**DETAILS: PIT & FLANGE,
SKID, RAFTER**

SCALE: AS NOTED
DATE: 2026-09-28
SHEET 12 OF 18

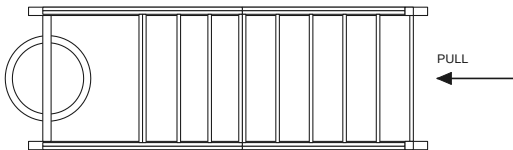
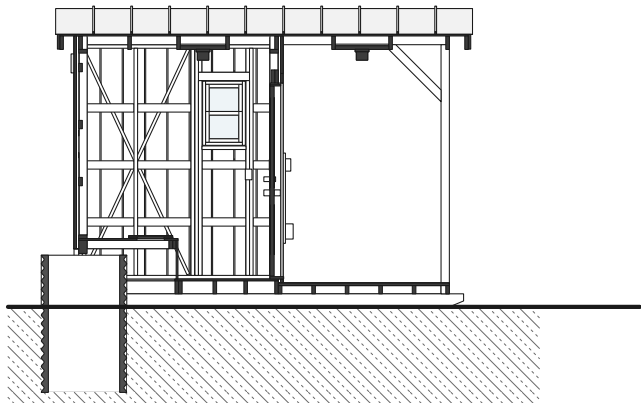
A-501

1 LAY OUT & EXCAVATE



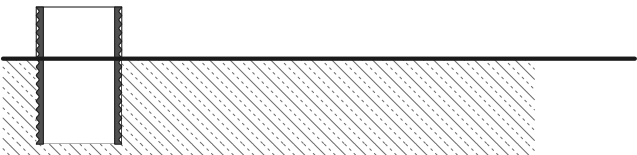
Clear and level the pad, crowned slightly so water runs away. Dig or auger a 44" hole 3'-0" deep; keep the bottom undisturbed.

4 SLIDE OVER THE PIPE



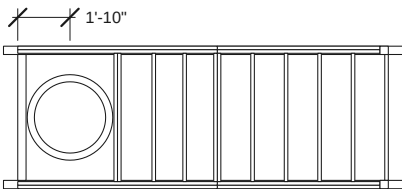
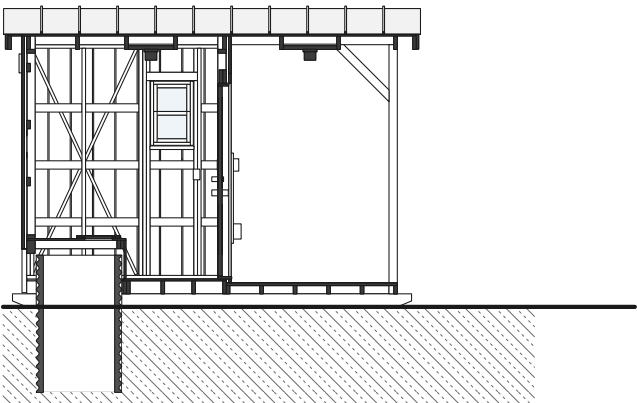
Pull on both skid ends with a bridle. The pipe passes between the skids (8 1/2" each side) and under the bench header (3/4" clear).

2 SET THE PIPE



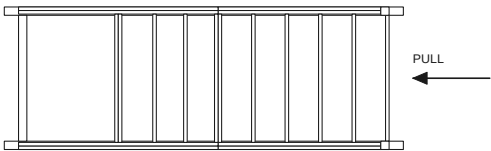
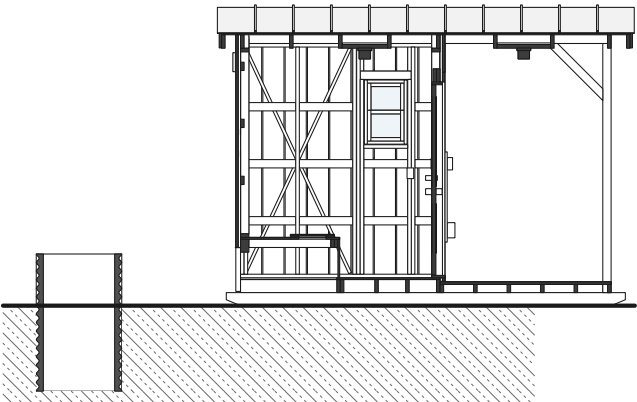
Cut the pipe to 4'-9 3/4". Set it plumb, top 21 3/4" above the skid path. Backfill in 6" lifts and tamp.

5 STOP AT THE MARK



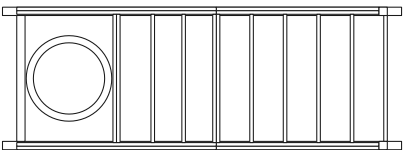
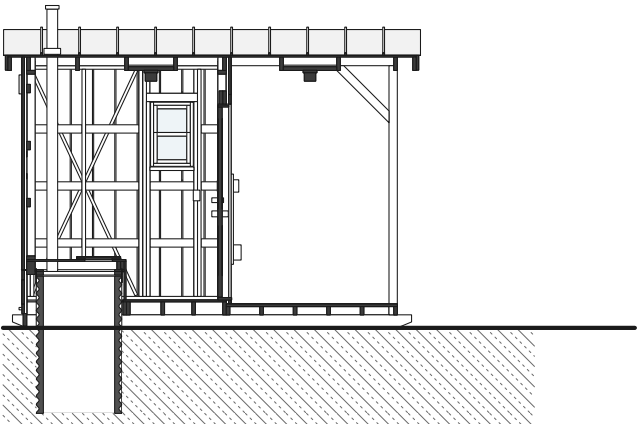
Stop with the pipe center 1'-10" from the outside of the back-wall framing (chalk a mark on each skid). Level the sled with PT shims under the skids.

3 STAGE THE PRIVY



Grade two skid paths level with the pad. Set the privy in line, back wall toward the pipe. Remove the back panel (10 lags) and the flange.

6 FLANGE, VENT, PANEL



Slide the flange plate + collar in from the back and screw it up to the bench framing; wrap and clamp the EPDM skirt. Fit the vent through the roof boot. Re-install the back panel.

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

SITE	Wooded lot, Rockingham County, NC (Piedmont)
OWNER	—
DESIGN	—

KEY DIMENSIONS	
Footprint (framing)	5'-0" x 7'-0"
Deck	5'-0" x 6'-0"
Skids	14'-0" long
Roof	3:12 single slope, metal
Pit	30" ID HDPE, 3'-0" deep
Height to roof	10'-5 9/16"

PARAMETRIC SOURCE
design/privy.json
model hash d81dcef27b
generated 2026-09-28

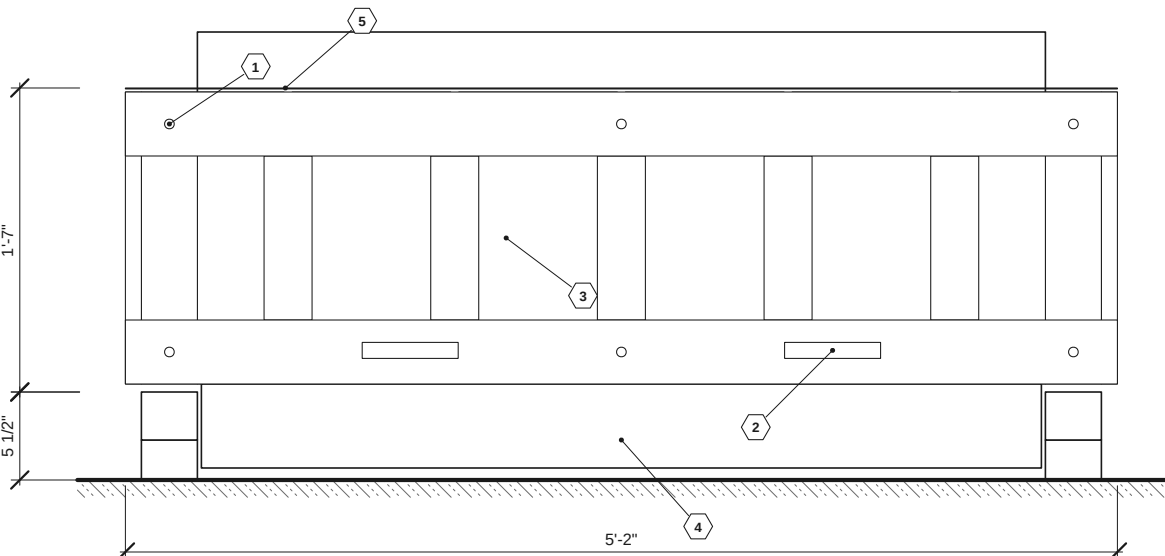
REV	DATE	DESCRIPTION
A	2026-09-28	Issued for review

INSTALLATION & RELOCATION SEQUENCE

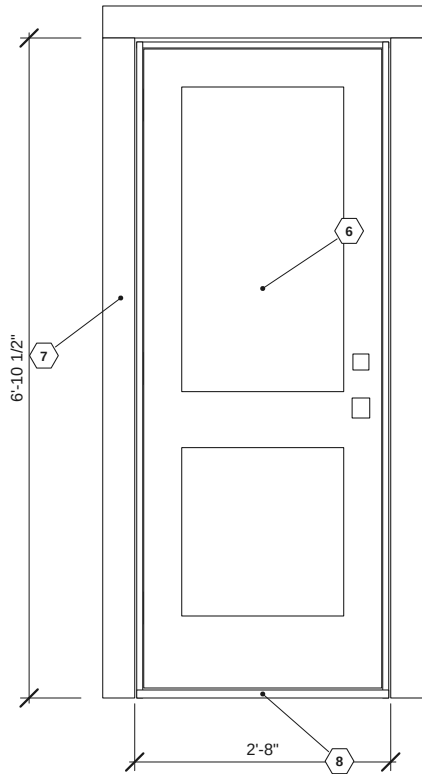
SCALE: AS NOTED
DATE: 2026-09-28
SHEET 13 OF 18

A-502

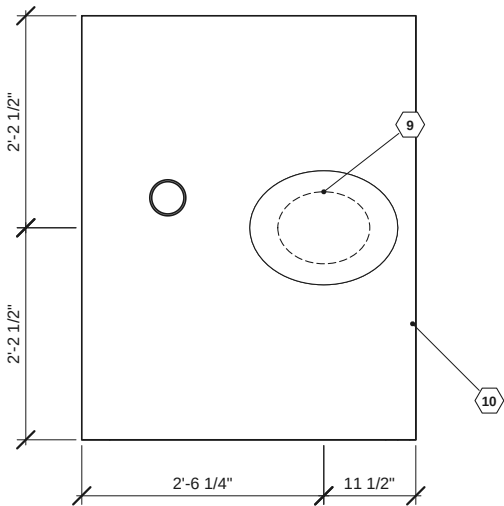
RELOCATION: reverse steps 6 to 3. Fill the old pit with at least 2' of clean soil, mound it and mark it. Pull the pipe for reuse or leave it and set a new one.



1 REMOVABLE BACK PANEL (FROM OUTSIDE)
SCALE: 1" = 1'-0"



2 DOOR D1 (EXTERIOR)
SCALE: 1/2" = 1'-0"



3 BENCH TOP (SEAT LID CLOSED)
SCALE: 1/2" = 1'-0"

KEYNOTES

- 1/4" x 3" galv. lag + fender washer: 3 per rail into corner studs and header, plus 2 into the rims (10 total)
- Galvanized pull handles
- 1x10 rough-sawn boards and 1x3 battens to match the walls
- PT 2x6 kick board between the skids (critter guard, part of the panel); untreated boards stop 6" above grade
- Z-flashing over the panel head
- D1: Fiberglass prehung exterior door, 2'-6" x 6'-8", in-swing, smooth 2-panel, 4-9/16" jamb. Keyed lever + deadbolt, factory hinges, adjustable sill, weatherstrip; exterior casing rough-sawn 1x4
- Rough-sawn 1x4 casing over the jamb; flash the head with Z-flashing and tape the jamb legs
- Adjustable sill on sill-pan flashing; 1 3/4" step down to the deck
- Seat hole 11 1/2" x 9", centered 11 1/2" back from the bench front, inside the pipe ID
- Bench front 3/4" plywood panel on 2x4 front rail

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

SITE	Wooded lot, Rockingham County, NC (Piedmont)
OWNER	—
DESIGN	—

KEY DIMENSIONS

Footprint (framing)	5'-0" x 7'-0"
Deck	5'-0" x 6'-0"
Skids	14'-0" long
Roof	3:12 single slope, metal
Pit	30" ID HDPE, 3'-0" deep
Height to roof	10'-5 9/16"

PARAMETRIC SOURCE

design/privy.json
model hash d81dcef27b
generated 2026-09-28

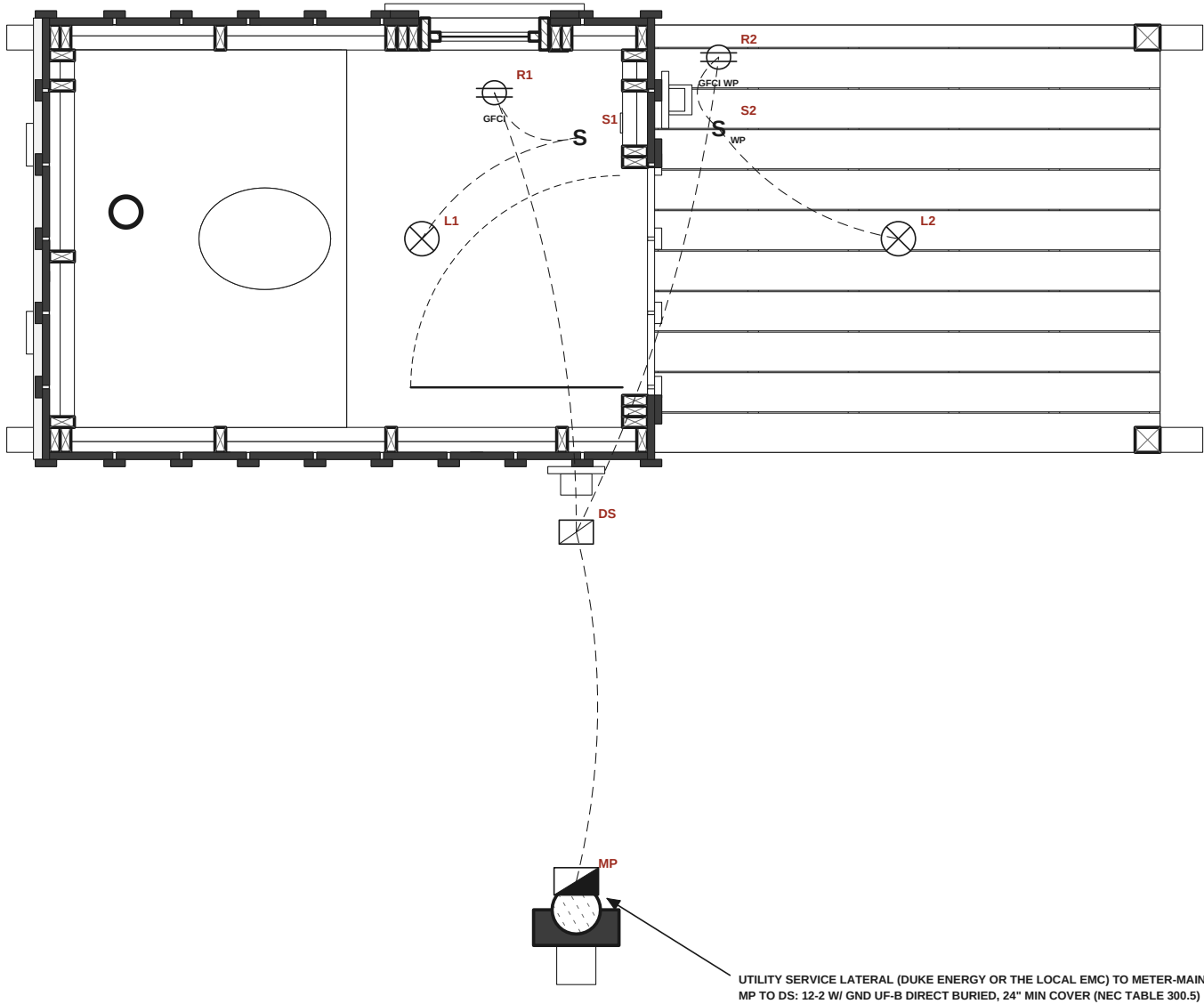
REV	DATE	DESCRIPTION
A	2026-09-28	Issued for review

SHEET TITLE

DETAILS: REMOVABLE PANEL, DOOR, BENCH

SCALE: AS NOTED
DATE: 2026-09-28
SHEET 14 OF 18

A-503



UTILITY SERVICE LATERAL (DUKE ENERGY OR THE LOCAL EMC) TO METER-MAIN MP
MP TO DS: 12-2 W/ GND UF-B DIRECT BURIED, 24" MIN COVER (NEC TABLE 300.5)

DEVICE SCHEDULE

TAG	DEVICE	MOUNT
MP	100 A meter-main combination, NEMA 3R, with branch spaces (utility-approved meter socket)	5'-6" above grade
DS	Weatherproof 20 A snap-switch disconnect at the point of entry (NEC 225.36); unhook point for relocation	3'-0" above grade
S1	Single-pole 20 A switch (interior light L1)	4'-0" AFF
R1	20 A tamper-resistant GFCI duplex	3'-8" AFF
L1	Damp-rated LED flush mount, about 800 lm, 3000-4000 K	ceiling
S2	Single-pole 20 A switch in weatherproof box / cover (deck light L2)	4'-0" AFF
R2	20 A WR + TR GFCI duplex, extra-duty in-use (bubble) cover	1'-8" AFF
L2	Damp-rated LED barn / jelly-jar light, about 800 lm, 3000 K	ceiling

SYMBOLS

	Meter-main / panelboard
	Disconnect switch
	Switch
	Switch, weatherproof
	Duplex receptacle, GFCI
	Ceiling light
	Circuit run (switch leg / feed)

ELECTRICAL NOTES

- Dedicated service: 100 A meter-main combination, NEMA 3R, with branch spaces (utility-approved meter socket) on a PT post 6'-0" off the low side. The utility service lateral, meter and main stay put when the privy moves. Coordinate the meter location and lateral route with the serving utility.
- Grounding at MP: (2) 5/8" x 8" copper-clad rods at least 6' apart, #6 Cu GEC, acorn clamps; neutral bonded only at the service disconnect (MP main).
- Branch circuit to the privy: 20 A GFCI breaker in MP, 12 AWG UF-B with ground (voltage drop on A-602). Spare breaker spaces for a later circuit.
- DS at the point of entry is the building disconnect (NEC 225.31/225.36) and the unhook point for relocation. No grounding electrode at the privy: single branch circuit with an EGC (NEC 250.32(A) exception).
- All receptacles GFCI (NEC 210.8); R2 weather- and tamper-resistant with an extra-duty in-use cover (406.9).
- UF-B with ground (damp location), stapled in stud bays; Sch 80 PVC where exposed below 8'. Unheated, damp building: NM-B is not permitted.
- Fixtures damp-rated. Keep 30 in x 36 in working space clear in front of MP (NEC 110.26).
- Permit and inspections: Rockingham County (NEC as adopted in NC). Utility releases the meter after inspection.
- Relocation: MP branch breaker off, open DS, disconnect the UF-B in the entry box, pull it from the conduit.

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

SITE	Wooded lot, Rockingham County, NC (Piedmont)
OWNER	—
DESIGN	—

KEY DIMENSIONS

Footprint (framing)	5'-0" x 7'-0"
Deck	5'-0" x 6'-0"
Skids	14'-0" long
Roof	3:12 single slope, metal
Pit	30" ID HDPE, 3'-0" deep
Height to roof	10'-5 9/16"

PARAMETRIC SOURCE

design/privy.json
model hash d81dcef27b
generated 2026-09-28

REV	DATE	DESCRIPTION
A	2026-09-28	Issued for review

SHEET TITLE

ELECTRICAL PLAN

SCALE: AS NOTED

DATE: 2026-09-28

SHEET 15 OF 18

E-101

1 ELECTRICAL PLAN

SCALE: 1/2" = 1'-0"

CUT LIST (1 OF 2)

AREA	STOCK	PIECE	QTY	LENGTH	NOTE
foundation	4x6 PT	Skid	2	14'-0"	6" x 2.5" chamfer both ends; tow hole
floor	2x6 PT	Floor joist	5	4'-6"	
floor	2x6 PT	Rim joist	4	7'-0"	
deck	2x4 PT	Deck joist	5	4'-6"	
deck	2x4 PT	Deck joist	1	4'-5"	
deck	2x4 PT	Deck rim	4	5'-8 1/2"	
deck	4x4 PT	Knee brace	2	2'-10"	45 deg both ends
deck	4x4 PT	Post	1	8'-8 7/8"	bear on skid; galv. post base + (2) 1/2" through-bolts to rim
deck	4x4 PT	Post	1	7'-6 3/4"	bear on skid; galv. post base + (2) 1/2" through-bolts to rim
deck	5/4x6 PT	Deck board	2	6'-0"	notch at post
deck	5/4x6 PT	Deck board	8	6'-0"	
deck	5/4x6 PT	Deck board	1	6'-0"	rip to width; notch at post
bench	2x4	Bench front rail	2	4'-5"	
bench	2x4	Bench joist	2	2'-10 1/2"	
bench	2x4	Bench side ledger	2	2'-10 1/2"	
bench	4x6 PT	Bench header	1	4'-5"	hang from doubled corner studs w/ face-mount hangers
wall	2x4	Blocking	12	1'-10 1/2"	
wall	2x4	Blocking	2	1'-9 3/4"	
wall	2x4	Blocking	6	1'-8 1/4"	
wall	2x4	Blocking	2	1'-6 1/2"	
wall	2x4	Blocking	3	9 3/4"	
wall	2x4	Blocking	3	9 1/8"	
wall	2x4	Blocking	4	7 3/4"	
wall	2x4	Blocking	7	2 3/4"	
wall	2x4	Cripple stud	1	6 5/8"	
wall	2x4	Header	2	1'-9 1/2"	
wall	2x4	Jack stud	2	6'-10"	
wall	2x4	Jack stud	2	6'-9"	
wall	2x4	King stud	1	8'-7/8"	
wall	2x4	King stud	2	7'-11 5/8"	
wall	2x4	King stud	1	7'-3 3/4"	
wall	2x4	Rough sill	1	1'-6 1/2"	
wall	2x4	Stud	1	8'-4 1/4"	
wall	2x4	Stud	1	8'-3 1/8"	
wall	2x4	Stud	5	7'-11 5/8"	
wall	2x4	Stud	1	7'-3 3/8"	
wall	2x4	Stud	1	6'-11 1/4"	
wall	2x4	Stud	1	6'-10 1/8"	
wall	2x4	Stud	6	6'-9 1/2"	
wall	2x4	Stud	1	6'-4 1/8"	
wall	2x4	Stud	1	5'-10 3/8"	
wall	2x4	Top plate	2	7'-0"	
wall	2x4	Top plate	2	4'-6 5/8"	3:12 slope
wall	2x4 PT	Bottom plate	2	7'-0"	
wall	2x4 PT	Bottom plate	1	4'-5"	
wall	2x4 PT	Bottom plate	1	1'-4 1/2"	
wall	2x4 PT	Bottom plate	1	4 1/2"	
wall	2x6	Header	2	2'-11"	
roof	1x6 rough-sawn	Barge board	2	7'-4 5/8"	
roof	1x6 rough-sawn	Fascia	2	14'-4"	
roof	2x6	Fly rafter	2	7'-2 5/8"	hung from sheathing + fascia; plumb-cut ends
roof	2x6	Frieze block	8	1'-7 1/8"	
roof	2x6	Rafter	8	7'-2 5/8"	3:12, birdsmouth both beams, plumb-cut tails

CUT LIST (2 OF 2)

AREA	STOCK	PIECE	QTY	LENGTH	NOTE
roof	4x6	Beam	2	13'-0"	
siding	1x10 rough-sawn	Siding board	1	9'-3 1/2"	
siding	1x10 rough-sawn	Siding board	8	9'-1 7/8"	
siding	1x10 rough-sawn	Siding board	1	9'-1"	
siding	1x10 rough-sawn	Siding board	1	8'-10 3/8"	
siding	1x10 rough-sawn	Siding board	1	8'-2 1/4"	
siding	1x10 rough-sawn	Siding board	1	8'-1 1/4"	
siding	1x10 rough-sawn	Siding board	1	7'-10 3/4"	
siding	1x10 rough-sawn	Siding board	9	7'-10 5/8"	
siding	1x10 rough-sawn	Siding board	1	7'-8 1/8"	
siding	1x10 rough-sawn	Siding board	1	7'-5 1/2"	
siding	1x10 rough-sawn	Siding board	1	7'-2 7/8"	
siding	1x10 rough-sawn	Siding board	1	7'-1/4"	
siding	1x10 rough-sawn	Siding board	3	5'-1 3/4"	
siding	1x10 rough-sawn	Siding board	1	1'-10 1/4"	
siding	1x10 rough-sawn	Siding board	3	1'-8 5/8"	
siding	1x10 rough-sawn	Siding board	1	1'-7 3/4"	
siding	1x10 rough-sawn	Siding board	1	1'-5 1/8"	
siding	1x10 rough-sawn	Siding board	1	1'-2 1/2"	
siding	1x10 rough-sawn	Siding board	4	1 1/2"	
siding	1x3 rough-sawn	Batten	8	9'-1 7/8"	
siding	1x3 rough-sawn	Batten	1	9'-1 3/8"	
siding	1x3 rough-sawn	Batten	1	7'-11 1/8"	
siding	1x3 rough-sawn	Batten	10	7'-10 5/8"	
siding	1x3 rough-sawn	Batten	1	7'-8 1/2"	
siding	1x3 rough-sawn	Batten	1	7'-5 7/8"	
siding	1x3 rough-sawn	Batten	1	7'-3 3/8"	
siding	1x3 rough-sawn	Batten	1	7'-3/4"	
siding	1x3 rough-sawn	Batten	4	4'-8 1/4"	
siding	1x3 rough-sawn	Batten	1	1'-6 3/4"	
siding	1x3 rough-sawn	Batten	1	1'-6 1/4"	
siding	1x3 rough-sawn	Batten	4	1'-4 5/8"	
siding	1x3 rough-sawn	Batten	1	1'-4 1/8"	
siding	1x3 rough-sawn	Batten	1	1'-1 5/8"	
siding	1x3 rough-sawn	Batten	1	11"	
siding	1x3 rough-sawn	Batten	1	10 1/4"	
siding	1x3 rough-sawn	Batten	6	1 1/2"	
siding	1x4 rough-sawn	Apron	1	2'-2 1/2"	
siding	1x4 rough-sawn	Head casing	1	3'-4"	
siding	1x4 rough-sawn	Head casing	1	2'-2 1/2"	
siding	1x4 rough-sawn	Side casing	2	6'-10 1/2"	
siding	1x4 rough-sawn	Side casing	2	2'-3 1/2"	
siding	1x4 rough-sawn	Sill	1	2'-4"	
removable panel	1x10 rough-sawn	Siding board	10	1'-6 1/4"	
removable panel	1x3 rough-sawn	Batten	5	10 1/4"	
removable panel	1x4 rough-sawn	Panel rail	2	5'-2"	
removable panel	2x6 PT	Kick board	1	4'-4 1/2"	rip to fit; part of the removable panel, closes the gap between the skids
other	1x8 rough-sawn	Mounting board	1	3'-2"	
other	1x8 rough-sawn	Mounting board	1	1'-4"	
other	2x6	Fixture block	1	1'-10 1/2"	2x6 flat between rafters for light fixture
other	2x6	Fixture block	1	1'-7 1/8"	2x6 flat between rafters for light fixture
other	6x6 PT	Service post	1	8'-6"	set in tamped gravel / concrete

PIT, ROOFING & HARDWARE (CONT.)

ITEM	QTY	NOTE
Round fixture boxes (pancake) + cable staples / connectors	1 lot	
8d ring-shank HDG siding nails	700	-2 per board per nailer
10d ring-shank HDG batten nails	300	through the gap, 16" o.c.
16d HDG framing nails / 3" structural screws	10 lb	
Exterior deck screws 2-1/2"	5 lb	decking and trim

LUMBER TO BUY

LUMBER	GRADE / TREAT.	LEN	QTY	BF
1x10 rough-sawn	Southern yellow pine, rough-sawn, air-dried	8'-0"	20	133
1x10 rough-sawn	Southern yellow pine, rough-sawn, air-dried	10'-0"	13	108
1x3 rough-sawn	Southern yellow pine, rough-sawn, air-dried	8'-0"	20	40
1x3 rough-sawn	Southern yellow pine, rough-sawn, air-dried	10'-0"	9	22
1x4 rough-sawn	Southern yellow pine, rough-sawn, air-dried	8'-0"	6	16
1x6 rough-sawn	Southern yellow pine, rough-sawn, air-dried	8'-0"	2	8
1x6 rough-sawn	Southern yellow pine, rough-sawn, air-dried	16'-0"	2	16
1x8 rough-sawn	Southern yellow pine, rough-sawn, air-dried	8'-0"	1	5
2x4	SPF #2	7'-8 5/8"	28	144
2x4	SPF #2	8'-0"	7	37
2x4	SPF #2	8'-8 5/8"	3	17
2x4 PT	PT UC4A GC	8'-0"	13	69
2x6	SPF #2	8'-0"	14	112
2x6 PT	PT UC4A GC	8'-0"	10	80
4x4 PT	PT UC4B GC	8'-0"	2	21
4x4 PT	PT UC4B GC	10'-0"	1	13
4x6	DF-L #2	14'-0"	2	56
4x6 PT	PT UC4B GC	8'-0"	1	16
4x6 PT	PT UC4B GC	14'-0"	2	56
5/4x6 PT	PT UC4A	8'-0"	11	55
6x6 PT	PT UC4B GC	10'-0"	1	30

Total 1057 board feet (nominal). Rough-sawn stock is sold by the board foot.

SHEET GOODS

SHEET GOODS / PIECES	QTY
3/4" PT plywood, CDX grade: Floor sheathing 5'-0" x 3'-7 1/2"; Floor sheathing strip 3'-4 1/2" x 3 1/2"; Floor sheathing strip 3'-4 1/2" x 3 1/2"	1
3/4" BC plywood, sealed: Bench top 4'-5" x 3'-5 3/4"; Bench front panel 4'-5" x 1'-4 1/4"	1
3/4" PT plywood: Flange plate 4'-5" x 3'-1 1/2"; Collar ring (from 2 half-rings) 3'-0" x 3'-0"; Collar ring (from 2 half-rings) 3'-0" x 3'-0"	2
5/8" CDX plywood roof deck: Roof sheathing 14'-4" x 7'-2"	4

PIT, ROOFING & HARDWARE

ITEM	QTY	NOTE
30" dual-wall corrugated HDPE pipe (ADS N-12 or equal), vertical, open bottom	4'-9 3/4"	cut square; top at 21 3/4" above grade
EPDM rubber strip, 45 mil	9'-11 1/8" x 6"	skirt around pipe / collar joint
Stainless worm-gear band clamp, 36"+ (or strapping)	2	over skirt, above and below joint
Closed-cell foam sill gasket	9'-5 1/8"	under flange collar
4" Sch 40 PVC (black) vent w/ screened cap	9'-4 7/16"	cut to fit; paint black
Roof pipe boot (EPDM, corrugated-metal type) for 4" pipe	1	
Vent cap / tee with stainless fly screen	1	
24 ga Galvalume snap-lock standing seam, 16" panels, 1" seams, over synthetic underlayment	11 @ 7'-7"	-110 sq ft; order cut to length; one piece save to eave
Synthetic high-temp roof underlayment	2 sq	over the plywood deck
Pancake-head #10 x 1" screws for panel nail strips	100	12" o.c. along each panel
Eave drip trim + low-eave closure	14'-7 1/2"	
High-side (shed ridge) trim w/ Z-closure	14'-7 1/2"	
Rake trim	15'-3/8"	both rakes
H-clips for 5/8" plywood	24	between rafters at unsupported edges
Z-flashing, galvanized	5'-2"	head of removable panel
Small single-glazed double-hung shed window, wood or vinyl, 18" x 27" unit	1	RO 18 1/2" x 27 1/2"
Fiberglass prehung exterior door, 2'-6" x 6'-8", in-swing, smooth 2-panel, 4-9/16" jamb	1	RO 32" x 82 1/2"; Keyed lever + deadbolt, factory hinges, adjustable sill, weatherstrip; exterior casing rough-sawn 1x4
Sill pan flashing + flashing tape	1	door rough opening
12" x 8" screened aluminum louver, high on the back wall (make-up air for the pit vent)	1	
Toilet seat, elongated, with lid	1	mount with stainless hinge bolts through bench top
Face-mount hangers for 4x6 (bench header to corner studs)	2	
Face-mount hangers for 2x4 (bench joists)	4	
Galv. 4x4 post base / standoff	2	fasten to skid
1/2" x 7" HDG carriage bolts w/ nuts & washers	4	posts to deck rim
1/2" x 8" HDG carriage bolts	4	beam to post (or post cap)
Let-in steel wall-bracing strap, 10'	6	
(10) 1/4" x 3" galv. hex lag screws w/ fender washers	1	panel to corner studs, header, rims
3/4" galv. shackle or clevis	4	tow points, both skid ends
100 A meter-main combination, NEMA 3R, with branch spaces (utility-approved meter socket)	1	MP; utility-approved socket
20 A 1-pole GFCI breaker (listed for the MP panel)	1	privy branch circuit
5/8" x 8' copper-clad ground rods + acorn clamps	2	at MP, 6' min apart
#6 bare Cu grounding electrode conductor	20 ft	MP to rods
Service entrance conduit, riser and fittings (per utility spec)	1 lot	lateral into MP
12-2 w/ ground UF-B cable (feed)	38 ft	MP to DS, direct buried
3/4" Sch 80 PVC conduit + sweeps, straps	14'-6"	feed risers at MP and DS
12-2 w/ ground UF-B cable (in building)	50 ft	DS to devices and lights
Weatherproof disconnect switch, 20 A, with box	1	DS
Weatherproof boxes + covers (switch / in-use receptacle)	2	S2, R2
Old-work / surface boxes + plates	2	S1, R1
20 A switches	2	S1, S2
20 A GFCI receptacles (TR; WR for exterior)	2	R1, R2
Damp-rated LED ceiling fixtures	2	L1 interior, L2 deck

(continued)

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

SITE	Wooded lot, Rockingham County, NC (Piedmont)
OWNER	—
DESIGN	—
KEY DIMENSIONS	
Footprint (framing)	5'-0" x 7'-0"
Deck	5'-0" x 6'-0"
Skids	14'-0" long
Roof	3:12 single slope, metal
Pit	30" ID HDPE, 3'-0" deep
Height to roof	10'-5 9/16"

PARAMETRIC SOURCE
design/privy.json
model hash d81dcelf27b
generated 2026-09-28

REV	DATE	DESCRIPTION
A	2026-09-28	Issued for review

SHEET TITLE
CUT LIST & MATERIAL SCHEDULES

SCALE: AS NOTED
DATE: 2026-09-28
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DESIGN CHECKS (COMPUTED FROM THE JSON)

CAT.	CHECK	VALUE	LIMIT	NOTE
Fit	Pipe passes between skids (clear each side)	8 1/2"	>= 1"	OK
Fit	Pipe passes between side-wall framing (each side)	8 1/2"	>= 1"	OK
Fit	Pipe top clears bench header while sliding	3/4"	>= 1/2"	OK
Fit	Pipe clears bench-front floor joist	1/2"	>= 1/4"	OK
Fit	Seat hole lies inside pipe ID (margin)	1"	>= 1/2"	info
Use	Clear floor in front of bench	3'-2 3/4"	>= 24" (IPC min 21")	OK knee room for a seated adult
Use	In-swing door clears bench when open 90 deg	8"	>= 2"	OK leaf folds against the hinge-side wall
Use	Standing room beside the swing (latch side)	16 1/2"	>= 12"	OK stand clear of the leaf while closing the door
Fit	Flange collar height (plate to pipe top less gap)	1 1/2"	>= 0	OK
Fit	Door header fits under sloped top plate	13/16"	>= 0	OK
Use	Door leaf clear height	6'-7 7/8"	>= 6'-0"	OK
Fit	Window header fits under top plate	10 1/8"	>= 0	OK
Durability	Untreated siding clearance to grade (IRC R317.1)	6"	>= 6"	OK PT skids, rims, plates and panel kick board are exempt
Use	Window sill above floor (privacy)	4'-8"	>= 4'-6"	OK
Use	Deck headroom under low beam	7'-2 1/4"	>= 6'-6"	OK
Use	Headroom at knee-brace foot (low side)	5'-2 1/4"	info	info braces are at the deck corners
Stock	Deck post length	8'-8 7/8"	<= 12'-0" stock	OK cut from 10'-0"
Stock	Longest stud	8'-4 3/16"	<= 10'-0" stock	OK cut from 8'-8 5/8"
Stock	Skid length	14'-0"	<= 16'-0"	OK
Framing	Rafter birdsmouth depth	7/8"	<= d/3 = 1 13/16"	OK
Framing	Vent located inside pit ID, clear of rafters/purlins/seat	found	found	OK
Framing	Let-in strap braces, left wall	2	>= 1	OK
Framing	Let-in strap braces, right wall	2	>= 1	OK
Framing	Let-in strap braces, front wall	0	>= 1	info wall too narrow for a 45-62 deg strap; racking taken by bench box, roof diaphragm and back wall
Framing	Let-in strap braces, back wall	2	>= 1	info
Structure	Rafter, snow + dead	fb/Fb' = 0.11; defl 0.009" (L/6199)	<= 1.0; <= L/180	OK 2x6 SPF #2, span 4'-8 1/2"
Structure	Roof sheathing span (APA 40/20 for 5/8", 32/16 for 1/2")	24"	<= 24" (5/8") / 16" (1/2")	info use H-clips at unsupported edges
Framing	Rake overhang carried by fly rafter + sheathing	8"	<= 12"	OK
Structure	Roof beam over deck	fb/Fb' = 0.13; defl 0.014" (L/5128)	<= 1.0; <= L/240	OK 4x6 DF-L #2, span 5'-10 1/4"
Structure	Floor joist	fb/Fb' = 0.22; defl 0.017" (L/3195)	<= 1.0; <= L/360	OK 2x6 PT SYP #2, span 4'-6"
Structure	Deck joist	fb/Fb' = 0.50; defl 0.066" (L/823)	<= 1.0; <= L/360	OK 2x4 PT SYP #2, span 4'-6"
Structure	Bench front rail, occupant at midspan	fb/Fb' = 0.99; defl 0.124" (L/427)	<= 1.0; <= L/360	OK 2x4 SPF #2, span 4'-5"
Structure	Bench joist, occupant at midspan	fb/Fb' = 0.67; defl 0.039" (L/926)	<= 1.0; <= L/360	OK 2x4 SPF #2, span 3'-0"
Structure	Bench header (back wall), occupant at midspan	fb/Fb' = 0.23; defl 0.014" (L/3870)	<= 1.0; <= L/360	OK 4x6 PT SYP #2, span 4'-5"
Pit	Usable pit volume (to 18" below grade)	55 gal (209 L)	info	info move the privy when solids reach the freeboard line
Pit	Estimated service life before relocation	12.7 yr	info	info 4 persons x 25 days/yr, 60 L/person-yr (dry pit, WHO range 40-90)
Electrical	Feeder voltage drop, 19 ft at 12 A	0.8% with 12 AWG Cu	<= 3%	OK use 12 AWG UF-B (20 A breaker); 12 AWG minimum
Electrical	GFCI on all receptacles (NEC 210.8)	GFCI breaker + GFCI devices	required	OK outdoor receptacle also WR with in-use cover (406.9)
Electrical	Disconnect at the building (NEC 225.31/36)	WP snap switch at entry	required	OK
Electrical	Service rating (NEC 230.79)	100 A meter-main	>= 60 A (230.79(D))	OK single small building; 100 A gives room for a later circuit
Structure	Skid soil bearing (dead + live)	605 psf	<= 1500 psf (IBC presumptive)	info
Moving	Estimated shipping weight (empty, no pit parts)	2,340 lb	info	info siding 515, roof 513, wall 319, deck 242, floor 161, door 150, foundation 147, site 125, bench 93, removable_panel 40, electrical 20, window 15
Moving	Towing force to drag on dirt (mu 0.5-0.7)	1,170-1,638 lb	info	info pull from both skid tow holes with a bridle
Loads	Roof snow load pf (ASCE 7: 0.7 Ce Ct Is pg)	12.0 psf	info	info pg=15 psf, Ce=1.0, Ct=1.2, Is=0.8

Structural rows are simplified NDS allowable-stress screening (No. 2 lumber, CD by load case, Cr 1.15 for repetitive members). Not a sealed design.

KEY PARAMETERS

PARAMETER	VALUE
building.width x depth	5'-0" x 7'-0"
building.low_bearing_height	7'-6"
deck.depth	6'-0"
roof.pitch, overhangs	3:12; eave 12", high 12", rakes 8" / 8"
pit.pipe ID / OD / bury	30" / 36" / 3'-0"
bench.height, hole setback	17", 11 1/2"
foundation.skids	skid on_edge, chamfer 6" x 2 1/2"
walls.stud_spacing	24"
roof.roofing	standing_seam on sheathing
pit.usage	4 persons x 25 days/yr
door R.O.	32" x 82 1/2", prehung, swing in
window unit, sill	18" x 27", 4'-8"

RENDER MATERIALS

	treated_lumber	Pressure-treated southern pine, lightly weathered
	framing_lumber	Kiln-dried SPF / DF framing lumber
	rough_siding	Rough-sawn southern yellow pine or yellow-poplar, al
	rough_trim	Rough-sawn pine trim
	decking	PT 5/4 deck boards
	treated_plywood	PT plywood
	bench_plywood	BC plywood, two coats exterior polyurethane
	galvanized	Galvalume (unpainted) standing-seam steel
	hdpe	Black HDPE corrugated pipe
	vent_pipe	Black PVC
	epdm	EPDM rubber
	steel	Galvanized / black steel hardware
	window_frame	Window frame and sash, white
	glass	glass
	seat	Toilet seat, molded wood / plastic, white
	void	Opening (crescent cut-out)
	soil	Packed brown dirt pad (slight red-clay cast), ~50% c
	forest_floor	Woodland floor: dense pine straw and brown oak leave
	pine_bark	Loblolly pine bark: platy, reddish-brown, dark fissu
	oak_bark	Post / white oak bark: grey, furrowed
	pine_needles	Loblolly pine foliage (needle-cluster blobs; texture
	oak_leaves	Oak foliage (leaf-cluster blobs; texture = per-clust
	door_fiberglass	Fiberglass door, factory painted deep green
	cdx_plywood	CDX plywood
	device	Switch / receptacle plate, ivory
	wp_box	Weatherproof box and cover, grey
	fixture	Light fixture body, black
	fixture_globe	Frosted fixture globe
	pvc_gray	Sch 80 PVC conduit, grey
	louver	Screened aluminum louver, bronze

LIGHTING

Sun azimuth 55 deg, elevation 38 deg; sky sunsky, turbidity 3.0; exposure 0.0 EV; tone map aces. Azimuths are measured in plan from the model +x axis (toward the deck) counter-clockwise toward +y (the high / window side). A view may override lighting (see renders.views[] lighting).

SKID-MOUNTED PIT PRIVY

Relocatable outhouse over a 30" HDPE pit, with covered deck

SITE	Wooded lot, Rockingham County, NC (Piedmont)
OWNER	—
DESIGN	—

KEY DIMENSIONS

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design/privy.json
model hash d81dcef27b
generated 2026-09-28

REV	DATE	DESCRIPTION
A	2026-09-28	Issued for review

SHEET TITLE

DESIGN CHECKS, PARAMETERS & RENDER SETUP

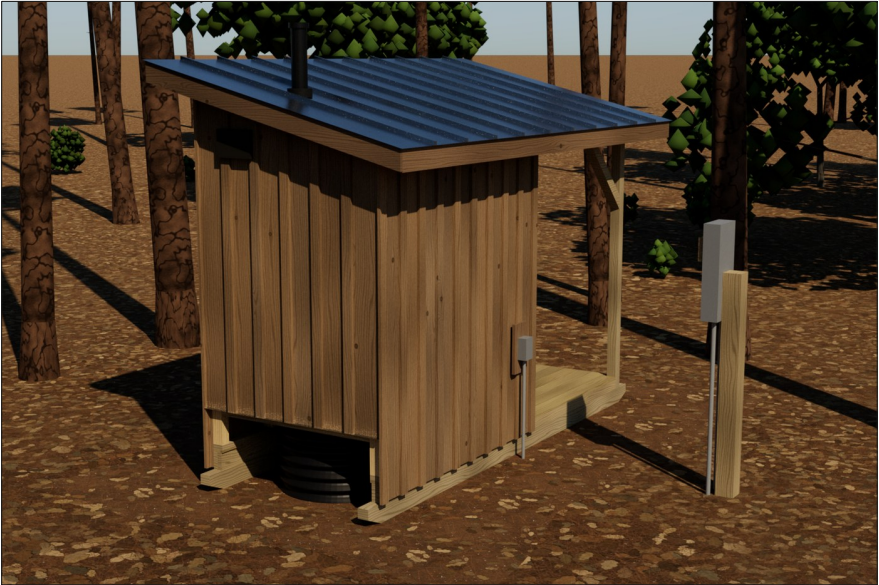
SCALE: AS NOTED
DATE: 2026-09-28
SHEET 17 OF 18

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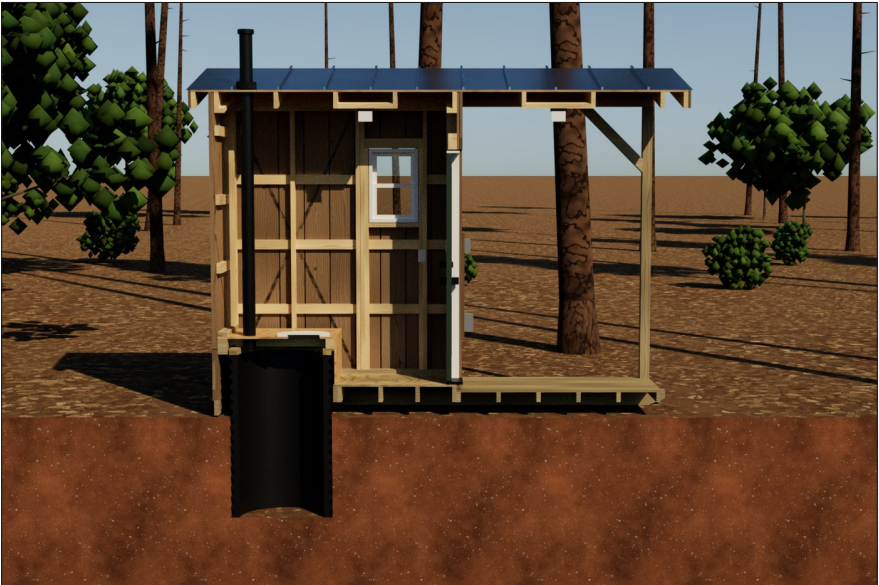


1 DECK AND WINDOW SIDE

Path traced with Mitsuba 3 from the same part model as the drawings. Materials, sun, sky, site and cameras come from the JSON (materials / lighting / renders).



2 BACK WITH THE REMOVABLE PANEL OFF: PIPE, FLANGE AND BENCH CAVITY (MORNING SUN)



3 CUTAWAY THROUGH THE PIT CENTERLINE

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RENDERINGS

SCALE: AS NOTED
DATE: 2026-09-28
SHEET 18 OF 18

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